MSci Simulation based modelling for decision making in healthcare

A MSci Project Report

by

Andrew Jones



School of Mathematics and Physics
Queen's University, Belfast

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Abstract

Background: Out-of-hospital cardiac arrests (OHCA) carry a high mortality rate, with survival critically dependent on time to defibrillation. Cairns et al. (2011) [1] assessed the cost-effectiveness of a mobile volunteer First Responder (FR) scheme in Northern Ireland (NI) using simulation-based modelling, concluding that it is highly unlikely to provide an efficient use of NHS resources in rural areas. Since then, the 2021 NI Census [2] and updated 2021 NI lifetables [3] indicate an ageing population. The National Institute for Health and Care Excellence (NICE) also revised its willingness-to-pay threshold to £30,000-£35,000 per QALY for new medicines in April 2026.

Aim: To investigate whether these developments alter the cost-effectiveness conclusion of Cairns et al. (2011), and to identify which FR allocation strategies and regions of NI offer the greatest benefit and cost-effectiveness. A secondary aim of this project is to investigate how ambulances queueing for handover to A&E affects response times and EMS effectiveness.

Methods: A Monte Carlo simulation model was developed following Cairns et al. (2011) [1] using a Poisson process for OHCA event generation, log-logistic AFT models for EMS and first responder response times, logistic regression for short-term survival, and a Markov microsimulation for long-term outcomes. Twelve FR allocation strategies were compared across all 26 NI Local Government Districts, with cost-effectiveness analysis conducted for Belfast, Carrickfergus and Fermanagh. A supporting Discrete Event simulation (DES) of the Northern Ireland Ambulance Service system was also developed.

Results: OHCA incidence weighted FR allocation outperformed all alternatives. Mean ICERs across 5-year simulations were £26,509/QALY for Belfast (95% CI £18,998-£37,637), £62,142/QALY for Carrickfergus (95% CI £21,776-£220,560), and £161,808/QALY for Fermanagh (95% CI £26,191-£817,373). The Belfast mean falls within the current NICE threshold of £30,000, with $\approx$ 70% of the 1000 simulations being cost-effective. For the DES study, it was found that reducing a single handover slot from a group of 5 increased mean handover time from 2.47 min (95% CI 2.29–2.64) to 13.64 min (95% CI 12.68–14.61).

Conclusion: The FR scheme is likely cost-effective under this model in Belfast under both the current and previous NICE thresholds, and is unlikely cost-effective in the most rural region of Fermanagh. Further work includes connecting the DES of NIAS with the Monte Carlo simulation for EMS and FR response.

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Chapter 1

Introduction

Healthcare systems worldwide face the challenge of allocating finite resources across a growing range of competing interventions. Increasing expenditure does not necessarily mean a proportional improvement in health outcomes [4], making the case for quantitative methods to support policy decisions. This situation is critical in Northern Ireland (NI), where the Northern Ireland ambulance service (NIAS) received 232,147 calls in 2024/2025 [5]. This has placed a huge strain on the service with ambulance handover times averaging 93 minutes in Belfast in December 2024 and only 7% of handover meeting the recommended 15-minute mark (Northern Ireland Audit office, 2025) [6].

Healthcare systems facing resource constraints often rely on quantitative modelling to support decisions about the allocation of limited resources across competing interventions, with simulation based modelling emerging as a particularly valuable tool in capturing stochastic complexity of health systems in ways observational data and analytical methods cannot [7]. Simulation based models are most useful when data for informing decision makers on new healthcare interventions is unobtainable, unethical or expensive to produce through trials. Monte Carlo simulation, Markov modelling, Discrete Event Simulation and agent-based modelling each offer their own strengths and weaknesses dependent on the study, and their integration into evaluation of healthcare systems have become increasingly common [8].

Cardiovascular disease remains one of the leading causes of premature mortality in the UK, with out-of-hospital cardiac arrest (OHCA) accounting for a substantial proportion of these deaths. Survival following OHCA is highly dependent on the time until defibrillation, with the probability of survival to hospital discharge declining by approximately 10% for every minute without treatment (Pell et al., 2001) [9]. Despite the large investment in emergency medical services (EMS), ambulance response time targets are frequently missed especially in rural regions where large geographies and limited station coverage creates unavoidable delays.

One proposed solution is the deployment of mobile volunteer First Responder (FR) schemes, in which community volunteers trained in basic CPR and the use of Automated

External Defibrillators are paged to respond alongside the Emergency Medical Services. An example of this is the Northern Ireland Public Access Defibrillation (NI-PAD) trial (Moore et al., 2008) [10], which provided early evidence of the potential of such schemes across urban and rural regions. Cairns et al. (2011) then developed a Monte Carlo simulation model to assess the cost-effectiveness of the NI-PAD scheme, concluding that it may not represent efficient use of NHS resources in rural Northern Ireland.

Since then, several developments suggest a potential improvement in the FR scheme's cost-effectiveness, with updated NI census data and life table data suggesting an older demography and a higher risk of OHCA, Dillenbeck et al. (2024) [11] providing more accurate long term outcome estimates, and NICE updating its willingness to pay threshold from £20K-£30K per Quality Adjusted Life Year in April 2026 [12][13].

This project will investigate whether the cost-effectiveness conclusions by Cairns et al. (2011) have changed in light of these developments. A Monte Carlo simulation model is developed incorporating these changes following the methods of Cairns et al. (2011), applied across all 26 NI Local Government Districts to compare twelve FR allocation strategies. A Discrete Event Simulation of the NIAS ambulance system is developed as a supporting component, and cost-effectiveness analysis is conducted for three contrasting case study regions (Fermanagh, Belfast, Carrickfergus).

Chapter 2

Literature review

2.1 Background and context of application example

Health care providers such as the NHS routinely face competing demands on limited budgets. Policy makers must decide which interventions (e.g new medicines or schemes) to fund, yet increasing spending does not always translate into proportional health gains [4]. The challenge of allocating resources efficiently across a wide range of possible interventions has caused an increasing interest in quantitative methods to support healthcare decision making.

Cairns et al. (2011) [1] demonstrate the use of this in the context of cardiac first responder (FR) schemes in Northern Ireland, noting that simulation based models enable decision makers to consider the impact of a scheme in different geographical regions than assuming the impact observed in one region will apply everywhere else. Their Monte Carlo simulation model (described in detail in Section 3) was built in response to observations from the Northern Ireland Public Access Defibrillation (NI-PAD) scheme to assess its effectiveness.

This project focuses on the application of simulation based modelling support healthcare decision making in NI, specifically through a Monte Carlo evaluation of a mobile volunteer First Responder scheme for out-of-hospital cardiac arrests (OHCA) in NI, and a Discrete Event simulation (DES) of the NI ambulance service.

2.1.1 Out-of-hospital Cardiac Arrests (OHCA)

Out-of-hospital cardiac arrest (OHCA) is a major cause of mortality in high-income countries, with over $30,000$ OHCAs occuring in the UK each year [14]. Data from the Resuscitation Council UK (RCUK) [15] show that return of spontaneous circulation (ROSC) is achieved in only around a quarter of OHCA cases, with 30 day survival rates reaching as low as 9.5%, 6.5%, and 9.6% in England, NI, and Scotland respectively.

Survival depends primarily on the chain of cardiopulmonary resuscitation (CPR):

- **Immediate recognition**: Møller et al. (2017) [16] concluded through an observational study using logistic regression of all OHCA patient emergency calls in Denmark that OHCA recognition by dispatch was positively associated with bystander CPR (odds ratio [OR] 7.84). They also reported the main predictors of OHCA recognition were addressing breathing (1.76 OR) and number of callers located by the patient's side (2.16 OR).
- **Early CPR**: The RCUK reports that 50.5% of OHCA are bystander witnessed, with high overall rate of bystander CPR (72.6%, 62.6%, 41.5% in England, Scotland and NI respectively). This early CPR can more than double survival chances with patients in the Utstein comparator group having a 30 day survival rate of 29.0% in England, compared to the overall 9.5% survival rate.
- **Rapid defibrillation**: Public Access Defibrillators (AEDs) use typically occur in fewer than 1 in 10 cases across the UK. In England, bystander AED use reached a record 9% in 2024 and approximately 11.5% in some regional reports.

Recent studies [17] of first-responder (FR) schemes indicate that smartphone based alert systems for volunteer responders increase early CPR rates from 55% to 63.8% compared to baseline, resulting in an improvement to long term survival from 9.4% to 14.4%. A Swedish study of urban and rural OHCAs by Berglund et al. (2022) [18] reported that FRs arrived with the shortest median response time of 6.0 minutes compared to EMS's 8.3 minutes, with a larger improvement in response time by the FR scheme observed in more rural areas.

However, social and regional inequalities persist as OHCA incidence is higher and bystander CPR rates lower in deprived urban communities [15]. Also incidence of OHCA is more likely to occur in regions with an older demography and higher population density. These factors demonstrate that implementation of a FR scheme in NI is likely to have a differing impact based on selected region demography and geography.

2.1.2 Northern Ireland Public Access Defibrillation (PAD) trial

The NI-PAD study (2004-2006) was a before and after trial designed to assess the effectiveness of a mobile volunteer First Responder scheme for OHCAs, who were equipped with Automated External Defibrillators (AEDs) [10]. The trial was conducted in the urban North and West Belfast (NWB) and the predominantly rural area encompassing Antrim, Ballymena and Magherafelt (ABM) district councils.

Volunteers were taken from the members of the public and the Police Service of Northern Ireland (PSNI), who were trained in AED use and dispatched alongside Emergency Medical Services (EMS) to OHCAs via an automated paging system [19]. In NWB 335 FRs were recruited and 29 AEDs deployed, and in the rural region 493 FRs were recruited and 53 AEDs deployed [10]. FRs were organised into a rotating weekly rota and carried their AEDs as they went about their daily lives, where they collected and deposited their AEDs each week.

Moore et al. (2008) [10] reported that during the intervention period FRs were paged to 128 OHCAs, attended 88 of these (68.8%), and arrived before the EMS in 18 of those 128 cases (14.1%). Most importantly the trial revealed differences in the scheme's performance between NWB and ABM, where EMS response times were faster and more

consistent in urban NWB while FR attendance rates and the proportion of cases reached before the EMS were higher in rural ABM. This demonstrated that the scheme likely has the greatest potential of benefit in rural areas where EMS response times are poorer. The trial also revealed a dispatch sensitivity (percentage of time the OHCA was correctly paged to FRs) of 60% in NWB compared with only 40% in ABM.

These findings from the NI-PAD trial demonstrated the complexity and geographical dependency of FR scheme performance, providing the empirical foundation for Monte Carlo simulations that followed by Marshall et al. (2006) [19] and Cairns et al. (2011) [1]. These sources are what the following work in this project are based upon.

2.2 Cost-effectiveness in healthcare

Healthcare systems face continuing pressure to adopt new technologies and treatments due to ongoing innovation and rising demand. However, decisions about allocating limited resources to these interventions present significant challenges relating to affordability, access, and opportunity cost for both patients and policy makers [8].

Cost-effectiveness analysis (CEA) provides a framework for evaluating whether new interventions deliver sufficient health benefits relative to their additional costs, compared with existing standards of care. As highlighted by Brandão et al. (2023), increased health-care spending does not necessarily result in proportional improvements in health outcomes, demonstrating the importance of economic evaluations such as CEA. [4].

In practice, CEA relies on a standardised measure of health benefit to enable comparison across interventions. Accordingly, the National Institute for Health and Care Excellence (NICE) recommends that CEA quantify health outcomes in terms of quality-adjusted life years (QALYs), as part of its health technology appraisal process (NICE, 2013, Section 5.3.1) [20].

QALYs combine duration and quality of life into a single metric by weighting each year of life by a utility score between 0 (death) and 1 (perfect health). This enables comparison of health benefits across interventions and disease areas and supports their use within simulation models.

The results of a cost-effectiveness analysis are commonly summarised using the incremental cost-effectiveness ratio (ICER), defined as the ratio of the difference in costs to the difference in health outcomes between two competing interventions

$$\text{ICER} = \frac{C_1 - C_0}{E_1 - E_0}, \tag{2.1}$$

where C_i and E_i denote the mean (expected) cost and health outcome (QALYs) of intervention i, respectively.

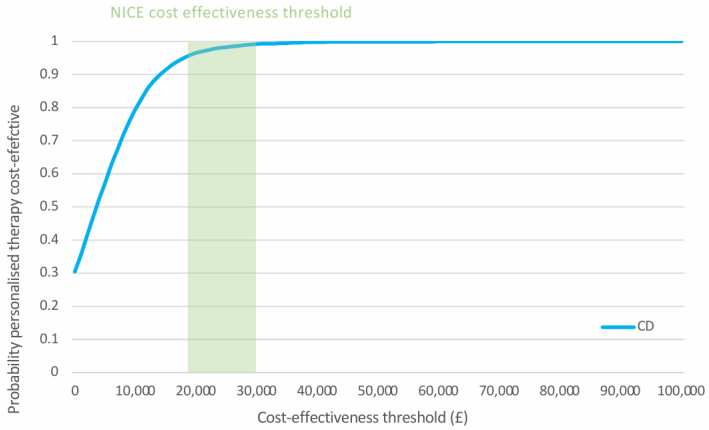
In practice the ICER is interpreted as the additional cost required to gain one additional QALY (cost per QALY gained), and is assessed against a willingness-to-pay (WTP) threshold representing the maximum amount a health-care system is prepared to pay for an additional QALY.

A practical evaluation must also quantify uncertainty [21]. In simulation-based CEA we

draw repeated, joint samples of incremental costs and effects (probabilistic sensitivity analysis, PSA) and summarise the resulting distribution with visual and decision tools. These tools include: the cost-effectiveness plane (a scatterplot of incremental cost vs incremental effect), cost-effectiveness acceptability curves (CEACs), and Net Monetary Benefit (NMB) summaries. These tools make uncertainty explicit and support probabilistic decision making rules e.g choose intervention by highest probability of being cost effective [22].



(a) Cost-effectiveness plane (5,000 PSA iterations) plotting incremental QALYs (x-axis) against incremental cost (y-axis). The dashed line represents a WTP threshold of £20,000/QALY; points below this line are considered cost-effective at this threshold.



(b) cost-effectiveness acceptability curve (CEAC) showing the probability the personalised strategy is cost-effective at different WTP thresholds (green band = NICE £20k-£30k/QALY).

Figure 2.1: Cost-effectiveness plane and CEAC reproduced from Buchanan et al. (2022), medRxiv preprint. Reproduced under CC-BY-NC-ND 4.0; figure used unchanged and attributed.

Buchanan et al. (2022) [23] provide a practical UK NHS example of a simulation-based CEA. A biomarker-guided personalised strategy for Crohn's disease is modelled using a decision-tree feeding a Markov model (14-day cycle, 20-year time horizon) from the NHS perspective with the base-case ICER reported as £2,176/QALY (Δcost £717; ΔQALY 0.330) . Figure 2.1 shows their 5,000 iteration PSA as both a cost-effectiveness plane and

a CEAC, where the CEAC indicates a $\geq 95\%$ probability the strategy is cost-effective at the NICE recommended £20k-£30k/QALY range. Buchanan et al. also present scenario analyses (e.g shorter time horizons or waning treatment effects) to demonstrate that point estimates must be interpreted in light of uncertainty and modelling assumptions.

NICE does not specify a single fixed numeric threshold, but its methods guide notes that interventions with ICERs roughly in the range of £20,000–30,000 per QALY are likely to be considered cost-effective, and that evidence supporting technologies with ICERs above £30,000/QALY must be increasingly strong (NICE, 2013, 6.3) [20]. In early April 2026, NICE implemented an increase in the standard cost-effectiveness threshold used in Technology Appraisals from £20,000–£30,000 to £25,000–£35,000 per QALY.

However, the Department of Health and Social Care (DHSC) has confirmed that this change only applies to NICE's Technology Appraisal guidance, where new technologies through this programme are primarily new medicines .[24, 25] NICE further states that, if a threshold change were agreed for the HealthTech and guidance programmes, it would follow a similar implementation approach there.[25]

Therefore an intervention such as a mobile first-responder scheme would not immediately follow a similar implementation unless in the government later agreed to extend the threshold change to HealthTech and guidance programmes.[24, 25]. This provides a basis for the potential use of a £35,000 threshold for an implemented NI volunteer FR scheme, as seen later in the study.

Empirical work suggests the best WTP threshold may be lower than the conventional NICE range. Claxton et al. (2015) estimate a central threshold of £12,936 per QALY and report a probability of 0.89 that the true threshold is below £20,000 and 0.97 that it is below £30,000 (Claxton et al., 2015; Scientific summary and Appendix 3, Table 183) [26]. This suggests that there is a divide in the literature surrounding the most beneficial NICE WTP threshold. These principles and decision rules form the basis for the simulation-based CEA developed in the subsequent chapters.

2.3 Simulation methods in healthcare decision making

Fone et al. (2003) [7] conducted a review of world literature from 1980 to 1999 and identified 182 papers that met the criteria for simulation modelling in healthcare spanning topics such as hospital scheduling, screening, economic evaluation etc. Their review concluded that the main power of simulation comes from being able to consider possible healthcare changes under safe and inexpensive conditions, and testing system limits (e.g Hospital bed capacity) without real world consequences. This is particularly relevant when observational data is limited or expensive/difficult to collect. Overall, this review highlights that simulation based modelling in healthcare has grown in popularity.

2.3.1 Markov models in Healthcare decision making

Markov modelling is a class of state-transition models widely used in health economic evaluation. Their use in medical decision making was formally established by Sonnenberg and Beck (1993) [27], who provided a practical framework demonstrating that Markov

models are particularly appropriate when clinical risk is ongoing, when the timing of events affects outcomes, and when costs and health effects must be accumulated across repeated cycles rather than assigned at a single terminal node. These properties make them well suited to chronic conditions (such as coronary heart disease) where patients face continuing risk of transitions between states such as survival, disability and death as shown in figure 2.2.

The origins of Markov models in health care dates before their widespread use in health economics. Beck and Pauker (1983) [28] were among the first to formally apply the discrete-time Markov process to clinical decision, where they demonstrated how transition probabilities between health states could be used to estimate expected survival and quality of life under competing clinical strategies

2.3.1.1 States, cycles and transition probabilities

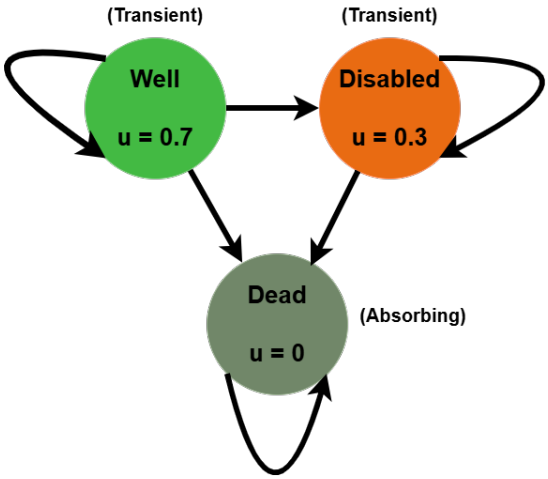


Figure 2.2: An illustrative Markov state transition diagram where each circle represents a discrete health state. Arrows indicate allowed transitions between states. Loops to self indicate that a patient may remain in the same state across a Markov cycle. Adapted from Sonnenberg and Beck (1993, Figure 3) [27], original redrawn.

In a Markov model, a patient population is partitioned into a set of mutually exclusive and exhaustive health states at each point in time. Time advances in discrete steps called cycles, and in each cycle a proportion of individuals in each state transition to another state according to fixed transition probabilities. A key feature of this is the Markov property which assumes that the probability of transitioning to the next state depends only on the current state and not on how the patient arrived to that state. As Briggs, Claxton and Sculpher (2006) [8] note, this memoryless assumption is a simplification and may not always hold clinically.

A state-transition modelling report by Siebert et al. (2012) [29] provides a reference for best practices in constructing state-transition models. This paper provides the methodological guidelines set for the Markov modelling of long-term outcomes in the present study (see Section 3.10). This paper also distinguishes between cohort-level Markov models and patient-level microsimulation. They define key requirements:

- A clear definition of the model aim and perspective.

- Cycle lengths and time horizons.
- The set of health states and their allowable transitions.

2.3.1.2 Cohort Models vs Markov Microsimulation

Markov cohort models track the proportion of a population in each state across cycles, computing expected costs and QALYs as weighted averages. However, these models are limited by Brennan et al. (2006) [30] provides a taxonomy of model types used for economic evaluation and identifies Markov microsimulations (Markov modelling at the individual level) as an extension that resolves this limitation by tracking each simulated patient separately through the model.

Hilligsmann et al. (2009) [31] demonstrates this approach in an economic evaluation of osteoporosis treatments, where they develop a Markov microsimulation that accounts for individual patient characteristics and calibrates the model to outputs observed by clinical data. This approach is adopted in the model presented in Section 3.10.

2.3.1.3 Markov modelling for cardiac arrest outcomes

The application of Markov modelling to OHCA survivors is motivated by the sustained excess mortality and ongoing care costs that survivors face after hospital discharge as described in Section 2.1.1. Cairns et al. (2011) [1] implement a Markov model for long term survival following OHCA in NI, structuring survivors into three neurological impairment states (no/mild, moderate, severe) derived from a retrospective cohort of Scottish SCA survivors by Pell et al. (2006) [32]. Transition probabilities combine age specific mortality from NI life tables with OHCA mortality derived from the AVID trial. The OHCA mortality is modified by whether the patient receives an ICD (relative risk 0.43 with ICD versus without). Patients with severe neurological impairment carry an additional relative risk of death of 1.1 (Nichol et al., 2003) [33].

2.3.2 Monte Carlo simulation for healthcare decision making

Monte Carlo simulation is used primarily in healthcare for Probabilistic Sensitivity Analysis (PSA), where model parameters are treated as random variables drawn from probability distributions rather than as fixed point estimates, and the model is run many times to determine parameter uncertainty. This is useful for decision makers as it quantifies the risk they take by implementing a new scheme or medicine.

Briggs, Sculpher and Buxton (1994) [21] were among the first to argue that uncertainty in the parameters of health economic models must be quantified and reported, instead of just addressing their affects through one-way sensitivity analysis. They argue that both stochastic uncertainty (which is the variability in each individual patients outcome) and the model uncertainty (uncertainty about the true values of model parameters) must be addressed and reported for a complete economic evaluation.

Claxton et al. (2005) [34] went further by arguing that PSA is not just optional for NICE technology appraisals but is a methodological requirement for if a new medicine is to be considered. They demonstrate that point estimate ICERs are often misleading when parameter uncertainty is large, and that decision makers need the full distribution of cost-effectiveness outcomes through techniques like the CE plane and the CEAC (see

Figures 2.1). This is demonstrated and implemented in the current model methodology section 3.11.

O'Hagen (2007) [35] states that a downside to implementing PSA with individual-level simulation is the high computational cost. They state that this situation often arises for patient-level simulation models (also known as micro-simulation models), where a single run of the model simulates thousands of patients. They also develop Analysis of Variance (ANOVA) based methods to reduce this high computational cost for Monte Carlo PSA models. In the current project this is relevant as the model shown in Section 3 (Figure 3.1 is highly computationally expensive to run, especially for one-way sensitivity analysis.

2.3.3 Agent based modelling (ABM) for healthcare decision making

Agent based models (ABM) are a class of bottom-up simulations in which a system is built from individual agents (e.g patients, clinicians, emergency responders) that follow rules and interact with each other and their environment to produce system level patterns [36]. A typical ABM specifies agents, their rules of interaction and environment to explore complex adaptive systems and feedback loops. For example, Leykum et al. (2012) [37] used an agent based model to show that variations in physicians' sense making abilities affected patient length of stay and complications, demonstrating that an ABM approach is suitable for exploring the impact of physician team behaviours on patient outcomes.

However, ABM is rarely used in healthcare studies, with a recent review [38] identifying just 31 ABM studies in healthcare. The Columbia Mailman School of Public Health notes that ABMS require detailed behavioural parameters that are hard to obtain and validate [36].

2.3.4 System Dynamics

System Dynamics (SD) is an approach to understanding the non-linear behaviour of complex systems over time developed in the late 1950s to study oscillations in industrial supply chains [39]. Its idea is that system behaviour over time is generated by internal feedback loops and not because of outside factors, and so policies should be designed by understanding the structure rather than reacting to symptoms [40].

2.3.4.1 Stocks, flows and feedback loops

An SD model is built from

- **Stocks**: Quantities that accumulate overtime, such as patients in A&E.
- **flows**: The rates that change stocks, such as the patient arrival rate or patient outflow rate
- **feedback loops**: Links stocks to the flows that change them. These are either classified as reinforcing (amplify change, e.g hospital congestion making staff burnout and worsening congestion) or balancing (counteracting change, e.g recruitment increasing as vacancy rates rise) [40].

Modellers usually start by drawing causal loop diagram to map out how everything connects, such as in figure A.1 demonstrating how A&E capacity can cause reduced response times. Causal loop diagrams are then turned into a stock-and-flow diagram that a computer can simulate qualitatively, such as in figure A.2.

2.3.4.2 System dynamics in healthcare

Homer and Hirsch (2006) [41] argue that SD is well suited to problems involving long time delays, accumulating risk factors and multiple feedback mechanisms in healthcare. They distinguish SD from microsimulation methods such as DES and ABM by noting SD tracks whole populations as a single stock level rather than tracking individual patients, making it more useful for strategic and policy questions. Recent systematic reviews [42, 43] identify patient flow, workforce demand, obesity and HIV/AIDS as the most common healthcare applications of SD, with steady growth since 2013.

One of the most relevant studies to this project is that of a London A&E department by Monefeldt and Rosenhead (2000) [44]. Their calibrated model showed that cutting hospital beds increased emergency admission waiting times modestly but sharply increased the cancellation of elective surgery. This showed that the effect of a resource shortage was transferred to a different patient group, illustrating Forrester's [45] claim that intuitive single issue policy fixes can have a counter-intuitive effects when feedback is ignored.

More recent SD applications have extended this approach to EMS, where Chen and Hung (2024) [46]used SD to model the surge capacity of Taipei regional EMS during the COVID-19 pandemic, simulating the dynamics of ventilators, physicians and nurses across 26 hospitals and using the model to test for future overloading events.

SD was not used for the present project as it combines populations into stocks rather than tracking individual characteristics, making it unsuitable when age, sex neurological state, location etc. determine the outcome as it does for OHCA survival. Also, standard SD is deterministic whereas this project requires probabilistic sensitivity analysis of incremental costs and QALYs [34]. However, it would be useful for use capturing the reinforcing loop between handover delays and ambulance availability in NIAS seen in this project's DES simulation and illustrated in figures A.1 and A.2.

2.3.5 Discrete event simulation

Discrete event simulation (DES) is a stochastic simulation methodology in which system state changes only when discrete events occur, widely used to model patient flow and resource contention in healthcare [47]. It was pioneered by Gordon at IBM with GPSS in 1961 [48] and by Tocher in the UK [49], and has since become one of the dominant operational research methodologies in healthcare.

DES has been applied in healthcare since the 1960s. Jun, Jacobson and Swisher (1999) [50] published the first widely cited survey, establishing DES as the standard methodology for modelling patient flow under resource constraints. Günal and Pidd (2010) [51] updated this,identifying emergency departments as the most studied setting and noting that most DES healthcare studies remained narrowly modelled to a single hospital or unit rather than whole systems. A more recent systematic review by Zhang (2018) [52] analysed over 200 healthcare DES studies from the 2010s and confirmed that emergency departments

remain the most common application. They state that more than one third of recent papers combine DES with other methods in hybrid models, however model validation and reporting transparency remain a weakness across the field.

Kotiadis et al. 2014 [53] argue that DES is suitable where entities (such as patients) queue for processes requiring resource utilisation and to capture variability in a system yet explore alternative scenarios. In healthcare simulations, it is common to include a DES when policy makers and analysts are interested in tails of a distribution and include individual effects in their model (such as queue prioritisation depending on individual characteristics) [54]. These principles explain why DES has become the dominant simulation methodology for emergency medical services, as ambulance systems combine call arrivals, priority queueing, stochastic service durations across job cycles and finite resources. All of these map directly onto DES's core structure.

DES applications to EMS go back several decades. For example, Ingolfsson et al. (2003) [55] developed a DES model to show that a single start (SS) station system for Edmonton EMS operations increased average ambulance availability and the fraction of calls reached within the department's response time standard. Lam et al. (2014) [56] utilised a DES model for Singapore EMS to evaluate the effectiveness of different dispatch policy alternatives, showing that modifications of the dispatch policy combined with the reallocation of existing ambulances reduced the gap between the ideal and existing response distribution by $11 - 13\%$.

2.4 Research Gaps

Several important changes have occurred since Cairns et al.'s 2011 study on simulating FRs in NI [1]:

1. **Demographic:** The 2021 NI Census [3] indicates an ageing population relative to 2001, which directly affects the expected annual incidence of OHCA in each area [57].
2. **Epidemiology:** The long-term survival component of Cairns et al.'s Markov model was parametrised using mortality data from the AVID trial [58] and NI life tables from 2004–06. Updated NI life tables from 2021 [59] now indicate decreasing mortality among males aged 70 and over, which is one of the most at risk demographic groups for OHCA. This change means that additional survivors from an FR scheme may now live longer on average than the original model assumed, potentially altering the QALY gains and cost-effectiveness conclusions.
3. **NICE updated guidelines** NICE implemented a change to its cost-effectiveness threshold in April 2026, raising the Technology Appraisal range from £20,000–£30,000 to £25,000–£35,000 per QALY [13]. These changes could alter the Cost-Effectiveness Analysis of a mobile FR scheme in Northern Ireland reported by Cairns et al (2011).
4. **First Responder optimisation:** Cairns et al. (2011) also only assessed the FR scheme in a single rural region of NI, and didn't investigate different ways of allocating FRs. This project extends this to all Local Government Districts (LGDs) across NI to find the most cost-effective LGD and way of distributing FRs based on their model.

Chapter 3

Methodology of simulation model

3.1 Study aim

The main aim of this project is to investigate the effectiveness of a mobile volunteer FR scheme across different regions of NI, and to see how the allocation of FR locations can affect the effectiveness of the scheme depending on the region. Another aim of this project is to see if the Cost-Effectiveness Analysis reported by Cairns et al. (2011) [1] has changed overtime with the introduction of a new NI Census indicating an ageing population [3], life-table data indicating a decrease in mortality among males aged 70 and over [60], and an updated NICE willingness to pay threshold shifting from £20k-£30k to £25k-£35k in April 2026 [12] [13].

3.2 Data Description

The model combines demographic and geographic inputs to simulate witnessed out of hospital Ventricular Fibrillation (VF) cardiac arrests in Northern Ireland to then assess the effectiveness and cost-effectiveness of a mobile volunteer First Responder (FR) scheme. The dataset used for these inputs is the 2021 NI census, where the dataset was built using the Northern Ireland Statistical Agency's (NISRA) flexible table builder selecting the variables age, gender and population for each Data Zone [2]. This data was used to estimate the expected annual number of cardiac arrests in each area for the Poisson process in section 3.4 and to assign age and sex characteristics to simulated events.

The parameters in the fitted Accelerated Failure Time (AFT) model (section 3.5) for ambulance response times were indexed by 1992 Local Government Districts (LGDs). Therefore a NISRA lookup table [61] was used to map each 2021 Data Zone to its corresponding 1992 LGD code and name. The same source supplied urban/rural classification data for 2021 Data Zones, which was used to model First Responder response times.

Data Zone boundaries (2021) were imported as a shapefile from NISRA [62] to create visualisations of the results from the model and to generate first responder home locations

within each assigned Data Zone polygon.

A separate R script was used to list NI ambulance stations, geocode, converted into spatial point data, checked manually where necessary and converted into Irish Grid coordinates along with Data Zone centroids. The resulting stations and locations were then checked manually according to named ambulance stations from a 2025 NI assembly answer [63]. A k-nearest-neighbours algorithm was then used to identify the nearest and second nearest ambulance stations for each Data Zone centroid, and calculate the distances between them for use in the AFT model for ambulance response time.

Finally, 2021 NI life tables were imported from the Office for National Statistics [59] to provide age and sex specific yearly mortality rates and remaining life expectancy estimates, which were used for calculating the long-term QALY and costs for simulated additional survivors gained by the FR scheme.

3.3 Algorithm overview

This algorithm follows the NI-PAD study [64] and Cairns et al. (2011) [1] study, but has been updated to reflect new geographic and demographic data, as well as updated FR schemes.

3.3.1 Sudden Cardiac Arrest generation

The model (Figure 3.1) was implemented as a Monte Carlo simulation in R, where each simulation run represents one year of witnessed VF sudden cardiac arrest (SCA) events across Northern Ireland. The overall aim of the algorithm shown in figure 3.1 was to compare an EMS only scenario with a simultaneous EMS + FR scheme, and then quantify both the effectiveness and cost-effectiveness of the addition of the FR scheme. The algorithm begins as follows:

1. For each 2021 Census Data Zone, the expected annual number of SCAs was calculated from its age-sex population structure and age-sex-specific SCA probabilities. Annual SCA counts were then generated using a Poisson process.

2. Each simulated SCA was assigned an age group, sex, hour of day, week of year, and geographic location for use in the subsequent response-time and survival models

3. An EMS response time was generated for each SCA using the fitted log-logistic accelerated failure time (AFT) model, with inputs based on the distances to the nearest and second-nearest ambulance stations together with time-of-day effects.

4. A yearly FR rota was then generated according to a chosen allocation rule. Volunteers were distributed geographically, assigned home locations within their allocated Data Zones, and scheduled into weekly rotating cohorts.

5. For each SCA, FRs within the paging radius were identified and then filtered through the dispatch, kit pick-up, and attendance components of the model. A response time was generated for each FR who responded.

6. The effective response time for each SCA was taken as the minimum of the EMS response time and all responding FR times. This best response time was then

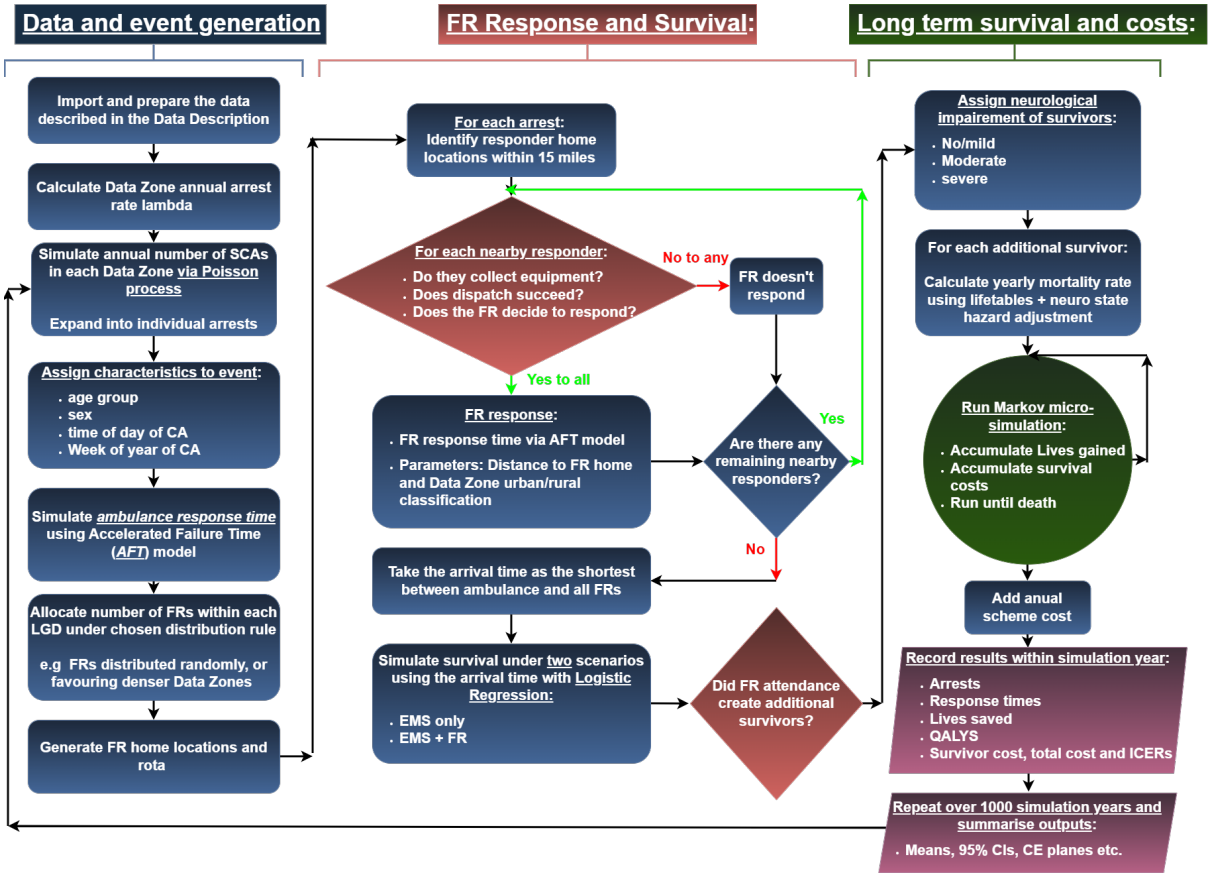


Figure 3.1: A flowchart of how the Monte Carlo simulation assesses the effectiveness and cost effectiveness volunteer First-Responder (FR) scheme. The blue squares represents processes, the red triangles represents decisions, the green circle represent the Markov micro-simulation for long term survival and cost, and the purple parallelograms represents the recording of data for the model results.

entered into the Valenzuela logistic survival model and compared with the EMS-only scenario to identify additional survivors attributable to the FR scheme.

7. Each additional survivor was assigned a neurological impairment state, together with the corresponding utility and cost inputs used in the long-term model.

8. These additional survivors then entered an annual-cycle Markov microsimulation, in which age and sex specific mortality from NI life tables was adjusted to reflect post-hospital discharge mortality after SCA. Within this microsimulation, survivors accumulated QALYs and long-term healthcare costs until death or the end of the model horizon.

9. Finally, yearly simulation outputs were combined into consecutive 5-year blocks so that total costs, QALYs, and incremental cost-effectiveness ratios (ICERs) could be calculated using the 5-year FR scheme cost. Repeating this process over many simulations produced Monte Carlo distributions of response times, short-term survival, long-term outcomes, and cost-effectiveness measures.

3.4 Cardiac arrest generation via Poisson Process

To simulate sudden cardiac arrest (SCA) incidence in a way that reflects the year to year variability observed in rare acute events, this model uses a Poisson Process to generate annual SCA counts within each Data Zone. This approach follows Cairns et al. [1], who construct an area-specific rate λ_i:

$$N_i \sim \text{Poisson}(\lambda_i), \tag{3.1}$$

$$\lambda_i = \sum_{j,k} P_{jk}\, n_{ijk}, \tag{3.2}$$

where P_jk is the probability of SCA over 1 year for persons in a given 5-year age-group j and gender k, and n_{ijk} is the number of people in Data Zone i from the 5-year age-group j and gender k.

These probabilities were derived using evidence based on the number of SCAs recorded in Belfast by the WHO Monica project data between 1985-1987 (Tunstall-Pedoe et al, 1994) [65], similar to Cairns et al. (2011) [1]. These probabilities were also informed by Moore et al. (2006) [64], where they reported the overall age-sex distribution of out-of-hospital SCA death cases in Belfast/Castlereagh (July 2003- August 2004), along with the proportions that were witnessed with a VF rhythm on EMS arrival [64].

Each simulated SCA is then assigned a 5-year age group and sex by sampling from its Data Zone's age-sex composition, weighted by the same age-sex probabilities used to estimate the expected number of SCAs. Each SCA is also assigned an hour of day based on evidence that out-of-hospital SCAs follow a morning peak and reduced overnight incidents, as reported by South East Coast Ambulance Service (2023-2024) [66] and Soo et al. (2000) [67]. Each SCA is also assigned a week in which they occur assumed to be uniformly distributed across the year, although there is evidence that OHCA events do vary by season with a peak in the winter months [68].

This approach assumes that the annual SCA counts are independent and follow the Poisson equidispersion property (i.e, the variance equals the mean). In settings where the empirical variablilty exceeds the Poisson expectation (overdispersion), a negative binomial specification can be explored as a sensitivity analysis [69].

Spatial variation in the Data Zone level rate parameters is shown in Figure 3.2.

3.5 EMS response via an Accelerated Failure Time model

Ambulance response time is an important determinant of outcomes following out-of-hospital cardiac arrest (OHCA). Pell et al. (2001) [9] reported that among OHCAs not witnessed by ambulance crews but attended within 15 minutes, 6% survived to hospital discharge. Using logistic regression, they estimated that reducing the 90th percentile response time to 8 minutes would increase predicted survival to 8%, and reducing it to 5 minutes would increase predicted survival to 10–11%.

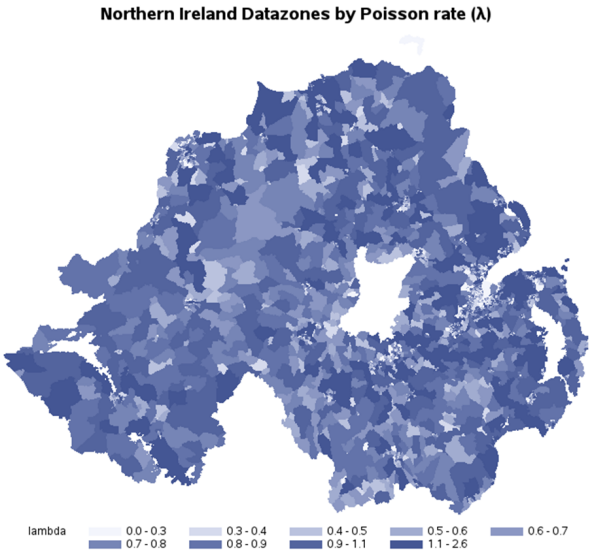


Figure 3.2: Calculated lambda rates for each NI Data Zone using Eq. 3.2 for each 2021 NI Census Data Zone.

Ambulance response time can be treated as a time-to-event variable, so a natural modelling framework is parametric survival analysis. In an accelerated failure-time (AFT) model, covariates act multiplicatively on the time scale (additively on $\log T$), so coefficients can be interpreted via time ratios with values > 1 lengthen response times and values < 1 shorten them [70].

Ambulance response times were simulated for each SCA using a log-logistic AFT model fitted by Cairns et al. (2011) [1] to 75,219 NI emergency response incidents. The log-logistic model provides a flexible parametric form that can capture right-skewed response-time distributions with occasional long delays, whereas simpler parametric choices such as Weibull impose more restrictive (monotone) hazard behaviour [70].

Let X_i denote the covariate vector for Data Zone and let β be the corresponding vector of estimated coefficients. X_i includes:

- $\log(r_{1i}+1)$, r_{1i} = Radial distance from Data Zone i's centroid to the closest ambulance station in miles.
- $\log(r_{2i}+1)$, r_{2i} = Radial distance from Data Zone i's centroid to the second closest ambulance station in miles.
- $I(r_{1i} < 2\,\text{miles})$ indicator.
- Indicator variables for the time of SCA occurrence for intervals: $[00:00-08:00)$, $[08:00-10:00)$, $[10:00-14:00)$, $[14:00-00:00)$, with $[14:00-00:00)$ as the reference category.

Under a log-logistic AFT model, the survival function for ambulance response time T_i conditional on covariates is

$$S(t \mid X_i) \;=\; \frac{1}{1+\left(\dfrac{t}{\exp(X_i^{\top}\beta)}\right)^{p}}, \tag{3.3}$$

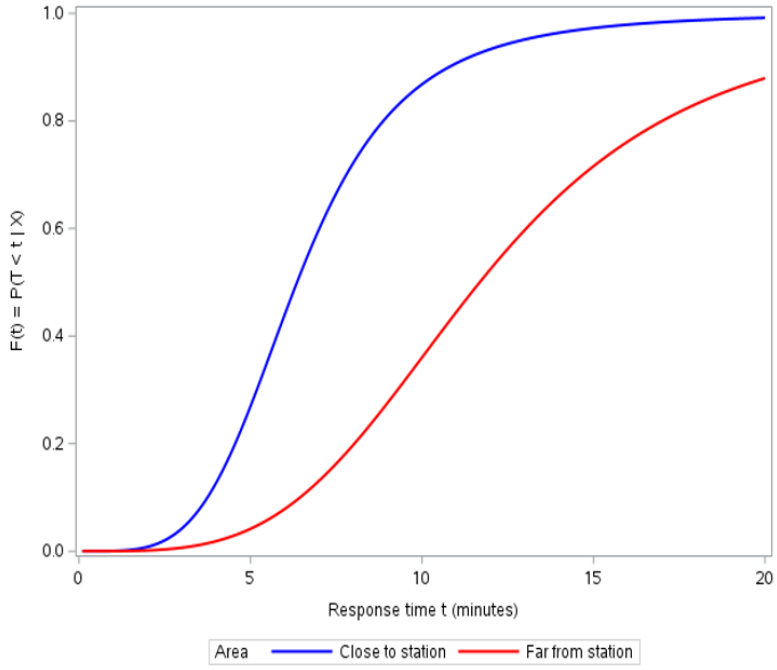


Figure 3.3: Cumulative distribution of ambulance response times in the hour block $10:00-13:59$ for an example Fermanagh Data Zone (blue) and an example Coalisland Data Zone (red). The Fermanagh Data Zone centroid is 2.67 miles and 20.1 from the nearest and second nearest ambulance stations and 20.1 miles from the second nearest ambulance station, and the Coalisland Data Zone is 4.1 miles and 7.4 miles from the nearest and second nearest ambulance stations.

where $p > 0$ is a shape parameter (in common AFT parameterisations, $p = 1/\sigma$) [71]. Response times are then generated by inverse-transform sampling: draw $S(T_i \mid X_i) \sim \text{Uniform}(0, 1)$ and solve for T_i

$$T_i \;=\; \exp(X_i^\top \beta) \left(\frac{1}{U} - 1 \right)^{1/p} \text{[71].} \tag{3.4}$$

Figure 3.3 shows cumulative distributions in which T_i is calculated from after sampling $S(T_i \mid X_i)$ for example Data Zones with differing distances from ambulance stations but at the same time of day, demonstrating how response times are skewed to the right with distance within this model.

3.6 Short-term survival via Logistic Regression

Step 6 in the simulation pipeline is converting simulated ambulance response times from the AFT model into survival-to-hospital discharge, where survival in this setting is a binary outcome (1 survived, 0 did not survive). Valenzuela et al. (1997) [72] develop a logistic regression survival model for witnessed out-of-hospital ventricular fibrillation (VF) arrests, using time intervals collapse-to-CPR (I_{CPR}) and collapse-to-defibrillation (I_{defib}).

Following Cairns et al. (2011) [1], the simulation treats short-term survival as primarily

driven by whether the arrest is witnessed with an initial VF rhythm, and then by the delays to CPR and defibrillation.

For cases simulated as witnessed VF, survival to hospital discharge is generated using the simplified Valenzuela logistic survival model based only on:

- I_{CPR}: interval (minutes) from collapse to CPR initiation,
- I_{defib}: interval (minutes) from collapse to defibrillation initiation.

Let t_{resp} denote the simulated response time (minutes). Then

$$I_{\text{defib}} = t_{\text{resp}} + 1, \qquad I_{\text{CPR}} = t_{\text{resp}} + 2. \tag{3.5}$$

where the additional 1–2 minutes represent practical on-scene delays to deliver the first intervention after arrival (positioning, attaching AED, initiating compressions).

Using coefficients reported for the simplified Valenzuela model, the survival probability is computed as

$$p_{\text{survive}} = \frac{1}{1 + \exp(0.260 + 0.106\, I_{\text{CPR}} + 0.139\, I_{\text{defib}})}. \tag{3.6}$$

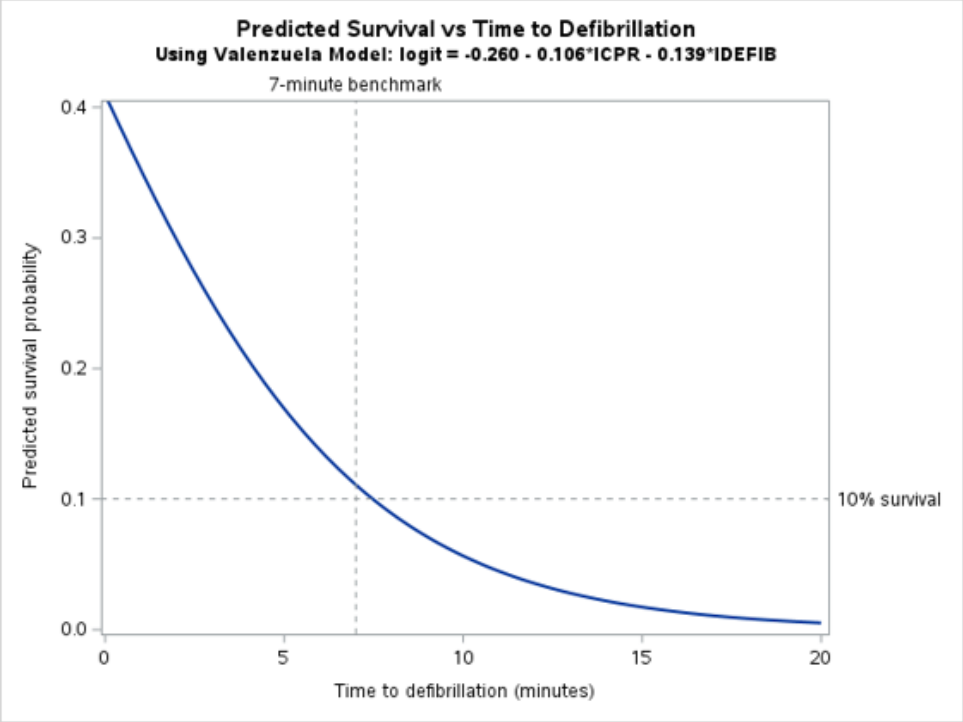


Figure 3.4: Predicted survival-to-discharge probability as a function of collapse-to-defibrillation time (I_{defib}) under the simplified Valenzuela et al. (1997) logistic survival model for witnessed VF out-of-hospital cardiac arrest.

Figure 3.4 shows that the predicted survival probability decreases rapidly as time to defibrillation increases. At collapse ($I_{\text{defib}} = 0$), the predicted probability is around 0.4, falling sharply to around 0.1 by $I_{\text{defib}} = 7$ minutes.

Finally, survival is simulated by drawing $U \sim \text{Uniform}(0, 1)$ and setting survival-to-discharge $= 1$ if $U < p_{\text{survive}}$ and 0 otherwise (i.e. a Bernoulli draw with probability p_{survive}).

In logistic regression, coefficients are interpreted on the log-odds scale. In the simplified Valenzuela model, a one-minute increase in I_{CPR} multiplies the odds of survival by $\exp(-0.106) \approx 0.90$, while a one-minute increase in I_{defib} multiplies the odds by $\exp(-0.139) \approx 0.87$. This implies that longer delays to CPR and defibrillation reduce the odds of survival multiplicatively with time.

3.7 First Responder allocation and rota generation

For each simulation, LGDs were assigned a FR rota which was distributed throughout the region according to a chosen allocation rule. These allocation rules shown in Table 3.1 were designed to compare different ways of distributing a fixed FR resource across Northern Ireland and the effect that they would have on the scheme. These allocation rules are treated as exploratory scenario rules rather than established formulae, and are used as simple demand and geography based experimentation motivated by the location-allocation literature [73, 74, 75]:

1. **Within-LGD random allocation (rule A6 in Table 3.1.** Each LGD was again assigned 31 responders, but these were distributed randomly across Data Zones within that LGD without reference to expected SCA incidence. This provided a benchmark against which the more structured allocation rules could be compared, and allowed the benefit of incidence based distribution to be assessed.

2. **Within-LGD allocation weighted by expected SCAs (rule A1 in Table 3.1.** Each LGD was assigned a 31 person FR rota allowing comparisons across LGDs under a fixed volunteer resource. These FRs were then distributed across the Data Zones within that LGD in proportion to expected annual SCAs. This rule therefore enforced equality between LGDs in the total number of responders received, while still prioritising higher risk Data Zones within each LGD. Allocation using the previous incidence multiplied by area rules were also ran for 31 FRs within each LGD to compare against this rule.

3. **NI-wide allocation weighted by expected SCAs. (rule B1 in the Table 3.1)**: All responders available across Northern Ireland were allocated in proportion to the expected annual number of SCAs λ_i in each Data Zone. Data Zones with a higher expectation of SCA therefore received more responders than areas with lower expectation of SCA. This rule prioritises epidemiological need and intuitively should represent the best allocation strategy.

4. **NI-wide and within-LGD allocation weighted mainly by expected SCAs, with a small area adjustment. (rules A2-A5, B2-B5 in Table 3.1** Two exploratory rules extended the previous approach by also accounting for geographic size. In these rules, each Data Zone received a score based mostly on λ_i, but with either a 10% or 20% contribution from Data Zone area. The aim was to test whether giving greater weight to the larger, usually underperforming areas (e.g Fermanagh) improved scheme performance across NI compared to the first rule.

5. **NI-wide allocation using incidence multiplied by area.** Further exploratory NI-wide rules combined expected annual SCA incidence and geographic area multiplicatively rather than additively. One rule was distributed weighted by $\lambda_i(\text{area in km}^2)^{0.25}$, while another $\lambda_i \log(1 + \text{area})$. These rules were designed to give additional priority to larger areas, but in a more moderate way than using raw area alone.

Rule	Scope	Allocation score
A1 Within-LGD incidence	31 FRs per LGD	λ_i
A2 Within-LGD 90/10 incidence-area	31 FRs per LGD	$0.9\,\lambda_i + 0.1\,a_i$
A3 Within-LGD 80/20 incidence-area	31 FRs per LGD	$0.8\,\lambda_i + 0.2\,a_i$
A4 Within-LGD incidence $\times$ area$^{0.25}$	31 FRs per LGD	$\lambda_i a_i^{0.25}$
A5 Within-LGD incidence $\times$ log-area	31 FRs per LGD	$\lambda_i \log(1 + a_i)$
A6 Within-LGD random	31 FRs per LGD	Equal/random
B1 NI-wide incidence	All NI	λ_i
B2 NI-wide 90/10 incidence-area	All NI	$0.9\,\lambda_i + 0.1\,a_i$
B3 NI-wide 80/20 incidence-area	All NI	$0.8\,\lambda_i + 0.2\,a_i$
B4 NI-wide incidence $\times$ area$^{0.25}$	All NI	$\lambda_i a_i^{0.25}$
B5 NI-wide incidence $\times$ log-area	All NI	$\lambda_i \log(1 + a_i)$
B6 NI-wide random	All NI	Equal/random

Table 3.1: First responder allocation rules compared in the simulation, with normalised expected SCA $\tilde{\lambda}_i = \lambda_i / \sum_j \lambda_j$, and normalised area/km^2 $\tilde{a}_i = a_i / \sum_j a_j$

3.8 First responder dispatch, attendance and response time

The resulting FR rota for each LGD was set to rotate four different cohorts of the same number of FRs every week. Home locations for each FR in the rota were then sampled randomly within each Data Zone. Each FR home address within 15 radial miles of a SCA were then selected/paged, and each FR responded based on passing these 3 main probabilities:

- Dispatch sensitivity: The rate at which SCAs were correctly identified by the dispatch mechanism in North West Belfast in 2004 was reported by Cairns et al. (2008) [76] to be 44.4% for VF out-of-hospital SCAs, and 47.1 for witnessed VF. If this fails, no FR responds in this model.

- The probability that a FR picks up their kit was reported to be 0.8 by the NI-PAD scheme [10] (double check this). If a FR doesn't pick up their kit that week, they are not able to respond to any nearby SCAs that occur during that week.

- The base probability that an individual FR is available to respond to a SCA after being paged is 0.3, as reported by Moore et al. (2006) [19] for the NI-PAD scheme. In this study, this base probability p_0 was assumed to decay with distance as a FR is less-likely to respond to far away SCAs with the relationship

$$p_{\text{FR}} = p_0 \exp(-0.1d), \tag{3.7}$$

where p_{FR} is the probability that an FR is available to respond and d is the radial distance of the FR from the SCA.

A response time was then calculated for each responding FR using another log-logistic AFT model fitted by Cairns. et al (2011) [1], with inputs:

- Natural logarithm of the radial distance of the FR from the SCA, $\ln(d)$.
- Urban/rural classification of the Data Zone in which the SCA occurs [61].

The same inverse-transform sampling procedure as for the EMS response (equation 3.4, Section 3.5) was applied to obtain these FR response times.

Next, the arrival time for each SCA was taken as the minimum response time between all responding FRs and the EMS response, which was then inputted into the same Logistic Regression model (Valenzuela 1997 [72]) described in section 3.6 to predict the short-term survival of each SCA. This was then compared with an EMS only scenario to simulate SCAs who would have survived if this FR scheme was implemented.

3.9 Health-state assignment, utilities and cost

Each survivor was assigned neurological impairment states based on probabilities from a retrospective cohort study of 1659 SCA survivors in Scotland (Pell et al., 2006) [32]. Next, impairment states were assigned utility values representing the quality of life for each patient within that state, shown in Table 3.2. Utility values for serious and moderate impairment were based on Gage et al.'s 1996 [77] study on utilities of stroke survivors, and the utility value for no/mild impairment is based on Nichol et al.'s 1999 study [78] on utilities of out-of-hospital SCA survivors.

Neurological state	Probability	Utility	Hospital cost (£)	Ongoing cost (£)
No/mild impairment	81.30%	0.8	34,540	6,168 per year
Moderate impairment	15.65%	0.2	13,040	398 per week
Severe impairment	3.05%	0.1	10,154	424 per week

Table 3.2: State specific long-term model inputs based on Cairns et al. (2011).

State-specific costs were then assigned using the inputs reported in Table 1 of Cairns et al. (2011). The 5-year implementation cost of the NI-PAD first-responder scheme (£474,256) was taken from a discussion paper about the FR scheme trial (West and Hamilton, 2007) [79]. Hospital costs for patients who died before discharge and for those discharged alive with no/mild, moderate, or severe neurological impairment were informed by Peberdy et al. (2003) [80], Department of Health (2006) [81], and Private Health Care UK (2007) [82]. Weekly long-term care costs for moderate and severe neurological impairment were taken from Curtis and Netten (2006) [83], while the annual ongoing cost of an Implantable Cardioverter Defibrillator (ICD) for the no/mild impairment state was taken from Owens et al. (1997) [84]. Future costs and QALYs were discounted at 3% per annum, following Gold et al. (1996) [85].

3.10 Long-term outcomes via Markov microsimulation

As Sonnenberg and Beck (1993) [27] note, Markov models are useful when clinical risk continues over time, when the timing of events affects outcomes and when costs and health effects (such as QALYs) must be accumulated across repeated cycles rather than assigned at a single terminal node. They also state that this approach is especially helpful when decision trees would require unrealistic simplifying assumptions to represent ongoing risk and long-term prognosis. Therefore long-term outcomes were modelled using an annual cycle Markov microsimulation in which each additional survivor was followed separately according to their age,sex and assigned neurological state. This is consistent with the Markov modelling for long-term survival implemented by Cairns et al. (2011), and is consistent with the broader literature such as its use in a 2009 cost-effectiveness study for treatments of Osteoporosis by Hiligsmann et al. [31].

Baseline annual survival rates were taken from the 2021 NI life tables so that mortality varied with the individuals age and sex over time. For each additional survivor, an exact starting age was sampled uniformly from within the simulated 5-year age band, and baseline annual survival probabilities were then updated according to age and sex.

In this current project's model long-term survival after hospital discharge was not parametrised using the AVID-based disease-specific mortality assumptions reported in their Table 3. Instead, to reflect the excess mortality experienced after cardiac arrest the model was calibrated so that survivors in the no or mild neurological impairment state matched the 5-year survival reported by Dillenbeck et al. (2024) [11] for patients discharged with Cerebral Performance Category (CPC) 1. Dillenbeck et al. reported a 5-year survival of 73.8% for CPC 1, with higher mortality among patients discharged with CPC 2 and CPC 3/4, corresponding to adjusted hazard ratios of 1.57 and 2.46 respectively relative to CPC 1. These estimates were then used to parametrise long-term mortality in the following Markov microsimulation, with no/mild, moderate and severe neurological impairment mapping to CPC 1, CPC 2 and CPC 3/4 respectively. External calibration of this type is a recognised approach in decision modelling when key parameters are not directly observable and model outputs must be matched to available empirical data [86].

Let $S_0(t)$ denote the baseline survivor function derived from the 2021 NI life tables. Under a Cox proportional hazards model [87],

$$h(t|x) = h_0(t)\exp(x^T\beta), \tag{3.8}$$

[71]

can be shown to imply

$$S(t|x) = [S_0(t)]^{\exp(x^T\beta)}. \tag{3.9}$$

This result shows that a group specific survival curve may be obtained by raising a baseline survival curve to the power of its hazard ratio (HR). In this model, the NI life table survival curve therefore acts as the baseline survival function, and the exponent term represents the mortality associated first with surviving a SCA and then with neurological

impairment severity.

For each additional survivor i, a 5 year baseline survival probability was first calculated from from the NI life tables using their sampled starting age and sex

$$S_{i,5}^{LT} = \prod_{k=0}^{4} S(a_i + k, \text{sex}_i), \tag{3.10}$$

where $S(a_i + k, \text{sex}_i)$ is the annual life table survival probability at age $a_i + k$ and sex. A SCA hazard multiplier h_{arrest} was then chosen so that the no/mild neurological impairment state reproduced the Dillenbeck et al. 5 year survival target of 0.738:

$$(S_{i,5}^{LT})^{h_{\text{arrest},i}} = 0.738. \tag{3.11}$$

Solving this yields

$$h_{\text{arrest},i} = \frac{\log(0.738)}{\log(S_{i,5}^{LT})} \tag{3.12}$$

This means that $h_{\text{arrest},i}$ is the amount by which the general population mortality must be increased so that a no/mild SCA survivor of that age and sex has the same 5 year survival as the CPC 1 group reported by Dillenbeck et al. (2024) [11].

The neurological state specific annual survival probability used in the Markov microsimulation was then

$$p_{surv,t}^{\text{state}} = S(a_t, \text{sex})^{h_{\text{arrest}} h_{\text{state}}}, \tag{3.13}$$

where

- $s(a_t, \text{sex})$ is the annual NI life table survival probability at age a_{it} and sex
- $h_{\text{arrest},i}$ is the calibrated SCA hazard multiplier
- h_{state} is the neurological state specific hazard ratio of the patient's state reported by Dillenbeck et al. (2024).

This calibration step was necessary as using NI life table data alone would imply that survivors with no/mild neurological impairment would have the same long term mortality as the general population of the same age and sex, likely overestimating long term survival of SCA survivors. However, this calibration assumes that Dillenbeck's CPC categories can be mapped to the model neurological impairment states, that the proportional hazards assumption holds, and that Dillenbeck's Swedish post discharge survival is transferable to this NI setting.

Each additional survivor then entered the microsimulation shown in Figure 3.5 in a fixed neurological state and was followed until death or the end of the model horizon. In each annual cycle, survivors could either remain alive in that state or transition to death. Annual survival in cycle t was modelled using equation 3.13.

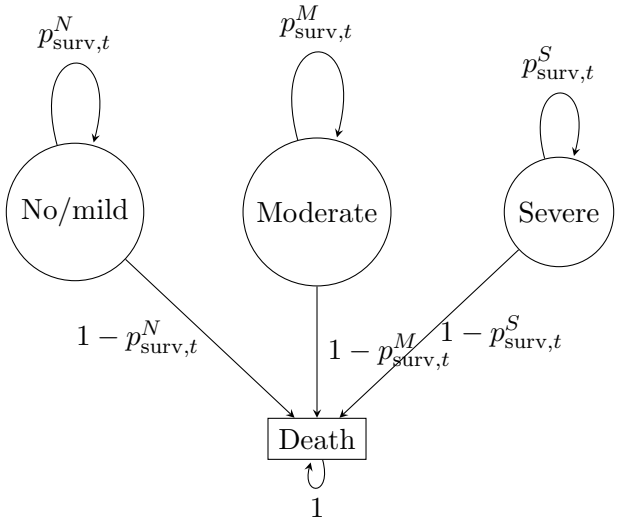


Figure 3.5: Markov transition diagram for the long-term microsimulation. Additional SCA survivors saved by the FR scheme enter the model in one of three fixed neurological states after hospital discharge. In each annual cycle they remain alive in that state or transition to death. Death is an absorbing state. $p^N_{\text{surv},t}$, $p^M_{\text{surv},t}$, $p^S_{\text{surv},t}$ are the no/mild, moderate, and severe neurological impairment annual survival probabilities respectively.

While alive, survivors accumulated their utility and cost associated with their assigned neurological state (Table 3.2) each year to calculate their total after discharge cost and total QALYs gained for the FR scheme. A half-cycle correction was applied so that individuals who died during a cycle received half a year of utility and half a year of ongoing cost, consistent with standard state-transition modelling practice [29].

3.11 Cost-Effectiveness Analysis

Cairns et al. (2011) [1] states that the main costs of the FR scheme include employment of a Community-Defibrillation-Officer to recruit, train, and co-ordinate volunteer FRs on a rota. Costs also include equipment, advertising and support administration costs over a 5-year period. Cairns et al. cites West and Hamilton 2007 [79] for the source of the 5 year FR scheme (£474, 256), and so was used as the 5 year FR scheme cost for 31 FRs per LGD.

Cost-effectiveness was then assessed by combining this FR scheme cost with the the long-term Markov microsimulation outputs. Although the model simulated outcomes for individual years, these yearly results were then grouped into consecutive 5-year blocks to match the 5-year FR scheme cost. For each 5 year block b, the total incremental cost of the scheme was calculated as

$$C_b = C^{\text{scheme}}_{5y} + C^{\text{survivor}}_b, \tag{3.14}$$

where C^{scheme}_s denotes the 5-year FR scheme cost and C^{survivor}_b is the total post-hospital discharge accumulated by the additional survivors caused by the FR scheme across the 5 simulated years within block b.

Similarly, the total incremental additional QALYs gained in each 5-year block were obtained by summing the QALYs accumulated by all additional survivors across the corresponding 5 simulated years. The total cost and additional QALYs gained were then discounted at a 3% rate according to Gold et al. (1996) [85]. The 5-year ICER for each block with QALY > 0 was then calculated as

$$\text{ICER}_b = \frac{C_b}{\text{Additional QALYs for block b}}. \tag{3.15}$$

For each LGD key outputs and results such as ICERs and QALYs were then summarised across 1000 simulated years to produce means and 95% confidence intervals to demonstrate uncertainty. Simulation-level cost and QALY outputs were also retained for presentation of uncertainty through CE planes and whether the simulated outcomes below varying willingness-to-pay thresholds.

3.12 Model validation

Before the effectiveness of the FR scheme is interpreted, the simulation outputs were compared against published empirical benchmarks to confirm that the model produces behaviour consistent with the real NIAS system and known OHCA epidemiology. First, the input parameters Poisson incidence rates [1], AFT response-time coefficients [1], Valenzuela survival logits [72] and Dillenbeck hazard ratios [11] are all taken directly from published empirical work. The Markov long-term component is explicitly calibrated to Dillenbeck et al. (2024) [11].Three main baseline EMS only statistics were validated: mean NI Response time, mean annual NI OHCA event counts, mean short-term survival to hospital discharge, and mean QALYs gained per life saved.

3.12.1 Ambulance response time

The simulated NI wide mean EMS only response time across 1,000 simulated years and all 26 LGDS, was 11.0 minutes (95% CI 10.68-11.30). With the FR scheme active, the mean time to first qualified responder dropped to 10.1 minutes (95% CI 9.81-10.37), a reduction of approximately 0.9 minutes. NIAS does not meet its Category 1 target of 8 minutes (mean) or 15 minutes (90th percentile) in any month of 2024/25 [88], and Unite reports Category 1 mean response times of 13-19 minutes during periods of acute service pressure [89, 90]. NIAS Trust Board papers for the period January-November 2025 report a mean Category 1 (cardiac arrest) response time of 12 minutes 16 seconds [91].

While slightly optimistic, the simulated mean still seems plausible compared to these figures suggesting the log-logistic AFT parametrisation from Cairns et al. (2011) [1] still produces response time distributions consistent with present day NIAS.

3.12.2 Annual OHCA event counts

The simulated model produced a mean of 2,627 witnessed-VF OHCAs per year (95% CI 2,529–2,726) for the NI population of 1,903,066, a simulated annual incidence of 138 per 100,000 (95% CI 133–143). This is roughly 1.6× higher than the figures reported

by the NI Department of Education ("around 1,500 OHCAs per year where resuscitation is attempted") [92] and the British Heart Foundation NI ("over 1,400 per year") [93].

This discrepancy likely arises because of the change in probability of OHCA P_{jk} (equation 3.2) derived from the Belfast WHO Monica project data 1985-1987 [65], which may be outdated. The model therefore operates at a somewhat higher OHCA density than the present-day NI baseline. Updating the Poisson rates to modern NIAS-registry data is flagged as future work in the Conclusion.

3.12.3 Short-term survival

The simulated EMS only survival rate for witnessed-VF arrests was 6.7% (95% CI 3.3%-10.4%). With the FR scheme active, simulated survival rose to 7.6% (95% CI 4.3%-11.1%), a lift of approximately 0.9 percentage points.

The Resuscitation Council UK reports NI's overall 30 day OHCA survival at approximately 6.5% [94], and the NI Department of Education and British Heart Foundation NI both cite "less than one in ten" as the headline survival figure for OHCA in NI [92, 93]. The simulated EMS only survival falls within 0.2% of the Resuscitation Council UK benchmark, providing strong validation that the Valenzuela survival logit [72] applied to the simulated response time distribution produces clinically plausible mortality outcomes.

3.12.4 Long-term QALY output

The model produced a mean of 4.79 QALYs per life saved (95% CI 0.30-11.57) under the A1 within-LGD incidence weighted allocation (rule A1), integrating over the 40-year Markov horizon and the neurological outcome states (no/mild, moderate, severe). For comparison, Cairns et al. (2011) [1] reported approximately 8.2 QALYs (95% CI 1.3–23.1) per life saved using a similar Markov model. This project's figure is lower than Cairns' due to the updated use of Dillenbeck et al.'s (2024) [11] hazard ratios, and the shorter time Markov time horizon of 40 years compared to Cairns' use of 70 years.

Chapter 4

Results

This chapter presents the results of the Monte Carlo simulation model described in Chapter 3, applied across the 26 Local Government Districts of NI. Three contrasting LGDs were selected as case studies: Belfast as a densely populated urban region, Fermanagh as the largest and most rural LGD, and Carrickfergus as an intermediate suburban region. Section 4.1 reports simulated response times under the EMS only and EMS + FR scenarios, Section 4.2 long term outcomes such as QALYs, Section 4.3 compares the twelve FR allocation rules defined in Table 3.1, Section 4.4 presents the cost-effectiveness analysis of the best performing rule and LGD (Belfast rule A1), Section 4.6 reports one way sensitivity analysis on the most influential parameters. All results are based on 1000 simulated years unless otherwise stated.

4.1 Response times

Figure 4.1 shows that simulated annual response times varied substantially across Fermanagh, Carrickfergus and Belfast reflecting differences in distance to ambulance stations and geography. Belfast had the shortest mean EMS response time at 8.2 minutes (95% CI 7.7–8.7 minutes), followed by Carrickfergus at 10.5 minutes (95% CI 9.0–12.3 minutes) and Fermanagh at 15.3 minutes (95% CI 13.3-17.7 minutes, nearly double Belfast's). These values are consistent with the geography of each region with Belfast having multiple ambulance stations within short distances of most data zone centroids, whereas Fermanagh's centroids are on average further from stations.

The addition of the FR scheme reduced mean response times in all three regions: to 7.7 minutes (95% CI 7.3–8.1 minutes) in Belfast, 9.3 minutes (95% CI 7.9–11.0 minutes) in Carrickfergus and 14.4 minutes (95% CI 12.6–16.6 minutes) in Fermanagh. This corresponds to reductions of 0.5, 1.2 and 0.9 minutes respectively. Notably, the response time reduction was largest in absolute terms in Carrickfergus rather than Fermanagh, despite Fermanagh having the longer baseline response times, reflecting the difficulty of achieving meaningful FR coverage in the most sparsely populated areas even with a 15-mile paging radius.

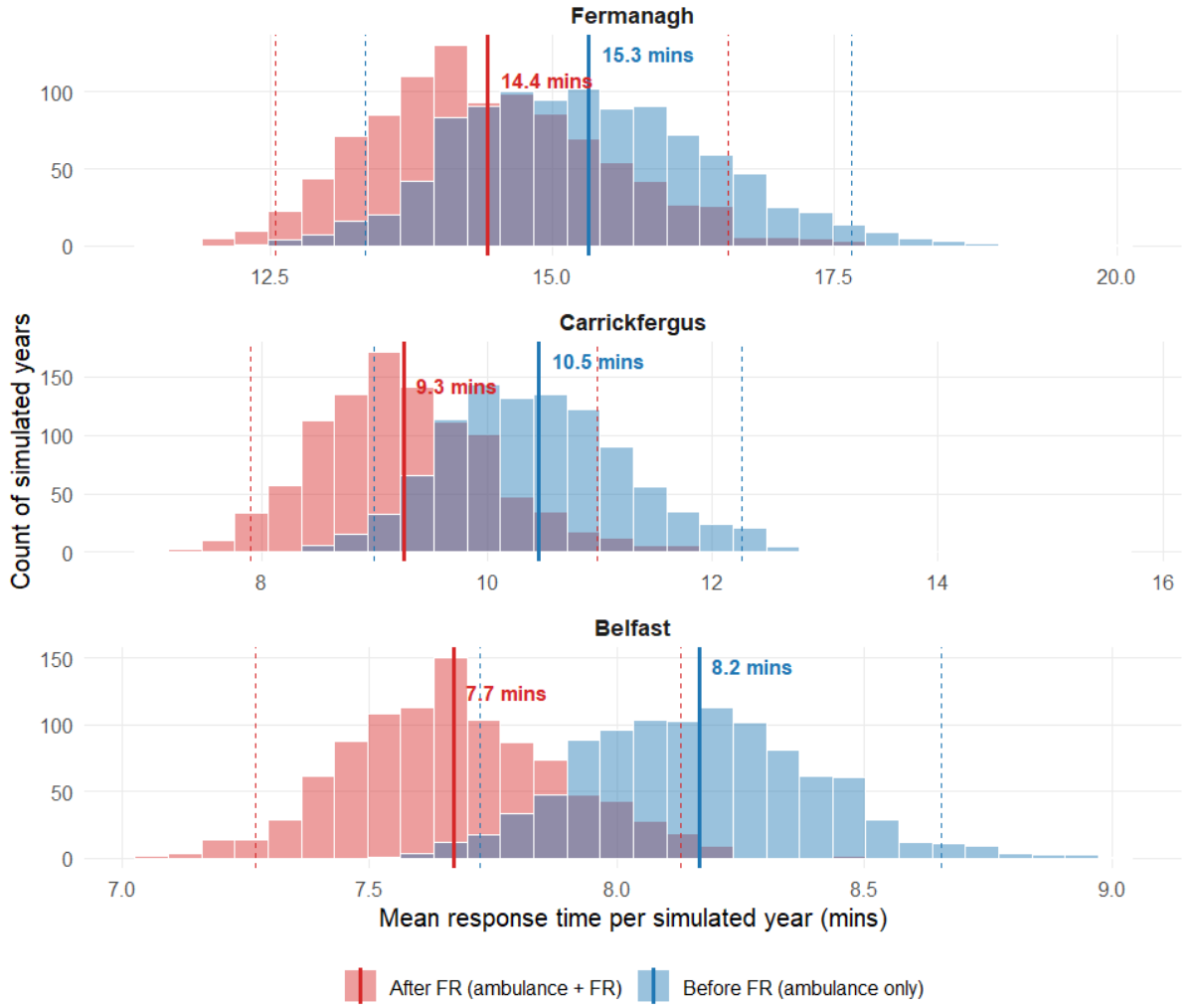


Figure 4.1: Distributions of the mean annual ambulance response time across 1,000 simulated years for Fermanagh, Carrickfergus and Belfast under the EMS-only scenario (blue) and the combined EMS and First Responder scenario (red). The FR allocation rule used was A1 in Table 3.1. Each panel uses a free x-axis scaled to the response time range of that region. Solid vertical lines indicate the mean and dashed vertical lines indicate 95% confidence intervals.

Figure 4.2 shows that these improvements in response time translated into a small but consistent increase in predicted survival across all three regions. Under the EMS-only scenario, mean annual survival rates were 9.1% in Belfast (95% CI 8.5–9.7%), 6.9% in Carrickfergus (95% CI 5.7–8.4%) and 4.8% in Fermanagh (95% CI 3.6–6.0%). The addition of the FR scheme increased these to 9.7% (95% CI 9.2–10.3%), 8.3% (95% CI 6.9–9.9%) and 5.3% (95% CI 4.1–6.6%) respectively. The absolute improvement in survival was largest in Carrickfergus (1.4 percentage points), followed by Belfast (0.6 percentage points) and Fermanagh (0.5 percentage points), which is consistent with the response time reductions achieved in each region.

Carrickfergus ran with the within-LGD incidence (A1) rule achieved the largest reduction in mean ambulance response time across all LGDs, which translated into the highest proportion of additional survivors attributable to the FR scheme compared to the EMS

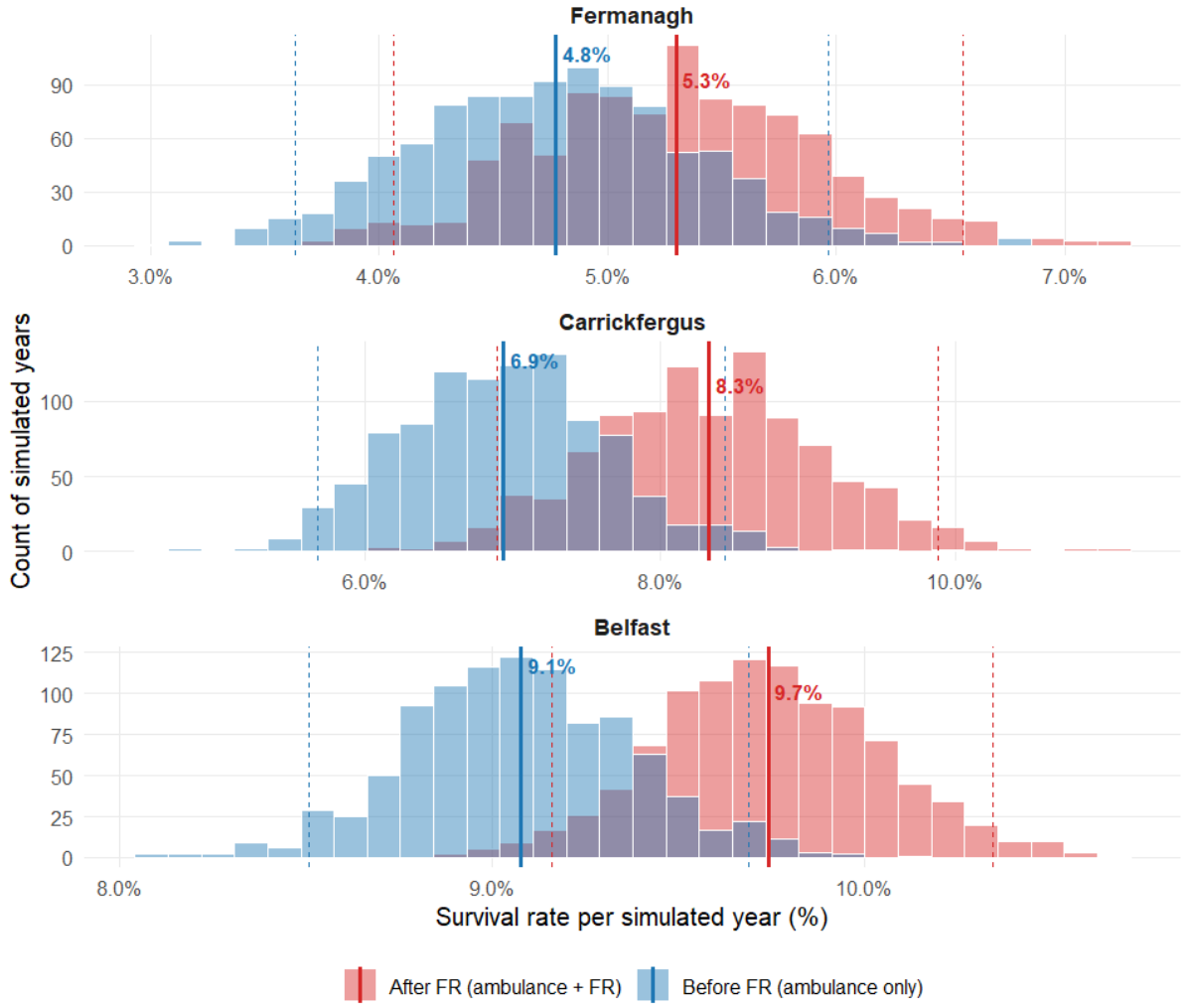


Figure 4.2: Distributions of the mean annual survival rates across 1,000 simulated years for Fermanagh, Carrickfergus and Belfast under the EMS-only scenario (blue) and the combined EMS and First Responder scenario (red). The FR allocation rule used was A1 in Table 3.1. Each panel uses a free x-axis scaled to the response time range of that region. Solid vertical lines indicate the mean and dashed vertical lines indicate 95% confidence intervals.

only scenario. Figure 4.3 shows that Carrickfergus has the highest mean proportion of lives saved by the addition of the FR scheme at 1.4% (95% CI 0.0%-4.9%), while Fermanagh and North Down had lower mean proportions at 0.5% (95% CI 0.0%-4.9%) and 0.7% (95% CI 0.0%-4.9%).

4.2 Long-term Outcomes

Across 1000 simulations the model predicted a mean of 2.6 additional survivors per year in Belfast (95% CI 0.0-6.0), 0.8 per year in Carrickfergus (95% CI 0.0-3.0) and 0.5 per year in Fermanagh (95% CI 0.0-2.0). The lower bounds of zero across all three regions reflect the fact that in many individual simulated years no additional survivors were generated by the FR scheme, particularly in Fermanagh and Carrickfergus where there where lower

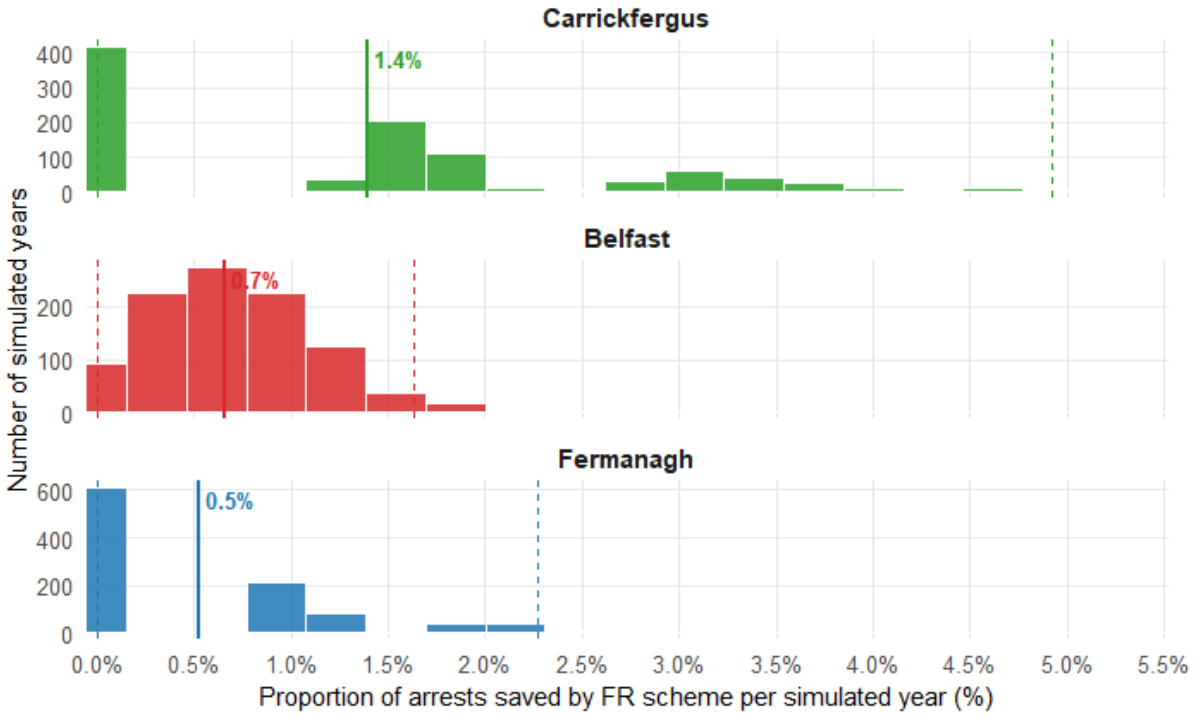


Figure 4.3: The distribution of the proportion of SCAs saved by the addition of the FR scheme for Carrickfergus (green), North Down (red) and Fermanagh (blue) across 1000 simulations. 95% confidence intervals are shown as vertical dotted lines, with the lower 97.5% quantile for all LGDs at 0%. The FR allocation rule used was A1 in Table 3.1.

OHCA rates.

Although Carrickfergus achieved the largest reduction in mean ambulance response time (1.2 minutes, Section 4.1), Belfast produced the most cost-effective FR scheme. This is because cost-effectiveness depends not only on the size of the response time improvement per arrest, but also on the absolute number of arrests to which that improvement applies. Belfast's higher OHCA incidence means that a smaller per arrest improvement accumulates into more additional survivors and QALYs per year, while the 5-year scheme cost is fixed regardless of region.

Mean QALYs gained per year from additional survivors entering the Markov microsimulation were 12.9 in Belfast (95% CI 0.0–33.5), 4.3 in Carrickfergus (95% CI 0.0–20.0) and 2.6 in Fermanagh (95% CI 0.0–16.4).

Figure 4.4 shows the relationship between QALYs gained and the number of FRs for Fermanagh and Carrickfergus. Both regions show slightly diminishing returns as FR count increases, with the rate of QALY gain flattening noticeably beyond approximately 60–70 responders in Carrickfergus and more gradually in Fermanagh. Figure A.4 shows this relationship extended to 500 FRs, demonstrating that Fermanagh overtakes Carrickfergus in QALYs gained at 350 FRs. These results suggest that increasing FRs beyond the baseline by even 60-70 could still yield meaningful additional QALY gains at relatively low marginal cost, especially for more rural Areas. Figure A.4 shows this relationship extended to 500 FRs, where Fermanagh overtakes Carrickfergus in QALYs gained at approximately 350 FRs. The continued linear rise by Fermanagh be beyond this point

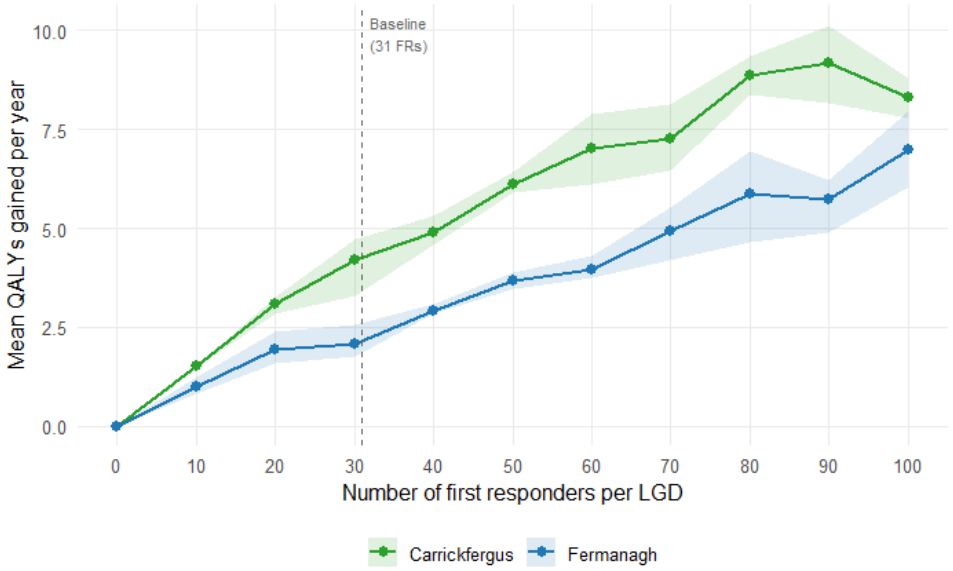


Figure 4.4: Mean QALYs gained per year by the FR scheme across varying numbers of FRs across Fermanagh (blue) and Carrickfergus (green). The FR allocation rule used was A1 in Table 3.1. Each point represents 3 seeds of 200 simulated years. Shaded regions represent 95% confidence intervals, and the dashed vertical line indicates the baseline scheme of 31 FRs as implemented by Cairns et al. (2011) [1].

reflects the slower saturation of a larger, sparsely covered region.

Figures 4.6 and 4.5 show that QALYs gained by the addition of 31 FRs distributed by incidence is highly dependent on the geographical size and population of a LGD. Belfast produced the highest mean QALYs per year by a considerable amount, consistent with its large population and high arrest incidence despite having one of the smallest geographies of any LGD. In contrast the largest LGD Fermanagh produced one of the lowest QALY gains, reflecting its sparse population and corresponding low number of OHCAs.

4.3 First responder allocation rules

Table 4.1: Within-LGD allocation rules (A1-A6) aggregated across all 26 LGDs, from the 12-rule × 26-LGD screening at 20 simulations per rule-LGD combination. ICER = total mean cost (with 5-year scheme cost annualised over 5 years) / total mean QALYs gained per year. Rules sorted by ICER.

Rule	Mean QALYs/yr	Mean Cost/yr	Mean EMS (min)	Mean EMS+FR (min)	ICER (£/QALY)
A1: Within-LGD Incidence	107.53	£4,475,626	10.95	10.05	£41,621
A3: Within-LGD 80% inc, 20% area	96.63	£4,081,475	10.95	10.12	£42,240
A6: Within-LGD Random	102.09	£4,325,872	10.95	10.04	£42,374
A2: Within-LGD 90% inc, 10% area	96.15	£4,455,120	10.95	10.05	£46,335
A4: Within-LGD Inc × area$^{0.25}$	86.51	£4,012,813	10.95	10.12	£46,388
A5: Within-LGD Inc × log-area	78.55	£3,975,224	10.95	10.14	£50,607

The performance of twelve alternative FR allocation rules across Northern Ireland under 20 screening simulations is summarised in Tables 4.1 and 4.2 for NI-wide and within-LGD rules. The distribution of each NI-wide (B) FR allocation rule is shown in the Appendix in figure A.3. Across both the within LGD allocations pure incidence weighting (rule A1,

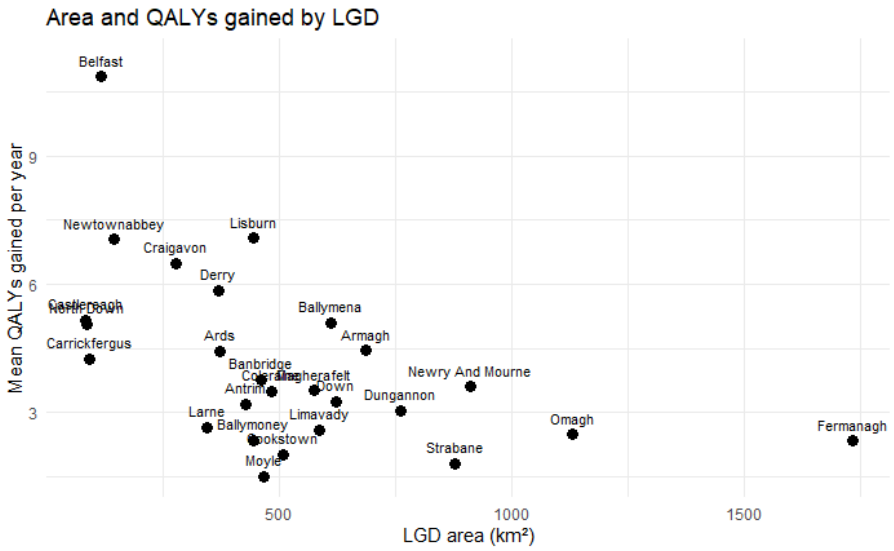


Figure 4.5: Mean Quality-Adjusted Life Years (QALYs) gained per year by the simulated FR scheme relative to area in km^2 across NI Local Government Districts. The FR allocation rule used was Within-LGD incidence (Table 3.1).

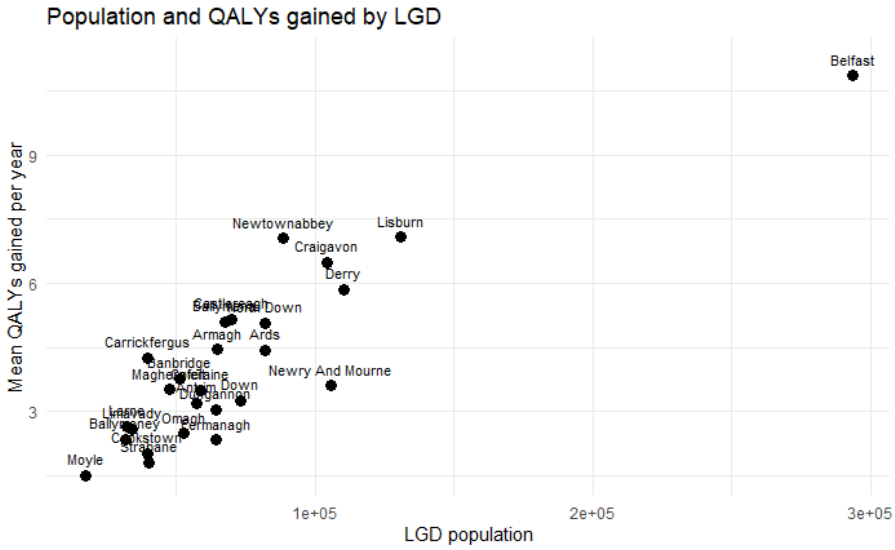


Figure 4.6: Mean Quality-Adjusted Life Years (QALYs) gained per year by the simulated FR scheme relative to population across NI Local Government Districts. The FR allocation rule used was Within-LGD incidence (Table 3.1).

Table 4.2: NI-wide allocation rules (B1-B6) aggregated across all 26 LGDs, from the 12-rule × 26-LGD screening at 20 simulations per rule-LGD combination. ICER = total mean cost (with 5-year scheme cost annualised over 5 years) / total mean QALYs gained per year. Rules sorted by ICER.

Rule	Mean QALYs/yr	Mean Cost/yr	Mean EMS (min)	Mean EMS+FR (min)	ICER (£/QALY)
B6: NI-wide Random	137.01	£4,893,103	11.78	10.86	£35,712
B1: NI-wide Incidence	130.74	£4,877,109	10.95	9.85	£37,305
B2: NI-wide 90% inc, 10% area	113.62	£4,511,260	10.95	9.91	£39,703
B3: NI-wide 80% inc, 20% area	102.93	£4,355,506	10.95	9.97	£42,315
B4: NI-wide Inc × $area^{0.25}$	88.77	£4,109,638	10.95	9.99	£46,297
B5: NI-wide Inc × log-area	69.90	£3,776,648	10.95	10.04	£54,028

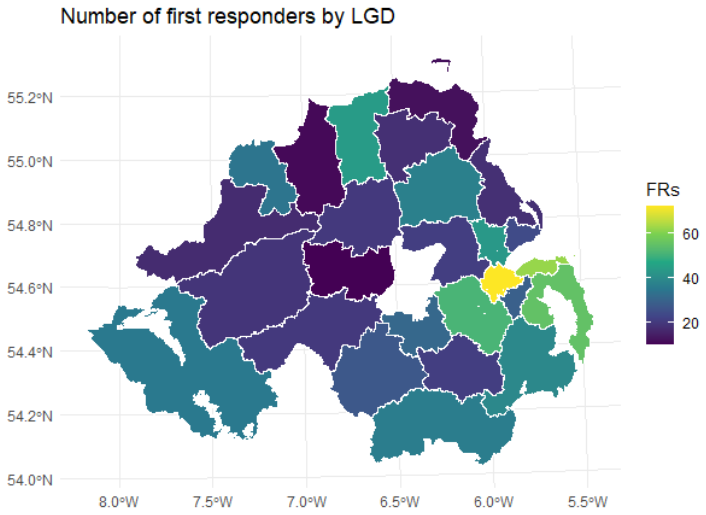


Figure 4.7: Distribution of FRs by LGD according to the NI-wide allocation rule of assigning FRs weighted by Data Zone specific expected annual number of SCAs λ_i.

Table 4.1) produced the highest mean QALYs per year and the lowest ICER, confirming that concentrating FRs in areas with the highest expected annual OHCA rates. However, Allocating FRs randomly across NI (rule B6) was the best allocation rule to use for the NI-wide rules (mean ICER £35,712), being marginally better than the pure incidence rule B1 (mean ICER £37,305).

The introduction of geographic area weighting consistently reduced QALY gains and increased ICERs The worst performing rule was incidence $\times$ log(area) rule, producing only 69.90 QALYs per year and an ICER of $£195,147$/QALY, suggesting that redistributing a set total of 806 (31 FRs for each of the 26 LGDs) FRs toward larger but sparser rural areas will perform poorly. This reflects a well known tension between efficiency and equity in allocating emergency resources, where strategies that maximise overall QALY gain can leave the most underserved rural populations with the poorest coverage [95].

Figure 4.7 shows the distribution of FRs across NI LGDs rule B1 in Table 3.1. Belfast received the largest allocation with 72 FRs, showing the trade off between allocating resources efficiently towards where arrests are most likely to occur but at the cost of reduced coverage to more rural areas.

4.4 Cost-effectiveness Analysis

The mean ICER from 5-year simulation blocks (incidence allocation rule) was £26,509/QALY for Belfast (95% CI £18,998-£37,637), £62,142/QALY for Carrickfergus (95% CI £21,776-£220,560) and £161,808/QALY for Fermanagh (95% CI £26,191-£817,373). Belfast's mean ICER falls within both the old and new NICE willingness-to-pay thresholds as discussed in Section 3.11, suggesting that the FR scheme represents a borderline cost-effective use of NHS resources. Fermanagh's mean ICER of £161,808 is well above the threshold, and the extremely wide CI (£26,191-£817,373) reflects the extreme variability in outcomes when annual OHCA counts are low.

These findings are somewhat consistent with those of Cairns et al. (2011) [1], who



Figure 4.8: Cost-effectiveness acceptability curves (CEACs) for Belfast with FRs distributed to Data Zones randomly (cyan) and FRs distributed weighted by expected number of annual SCAs for each Data Zone (gold). These curves are shown against previous (green) and updated (purple) NICE willingness-to-pay thresholds, £20k-£30k and £25k-£35k respectively.

reported that the rural region including Fermanagh, Cookstown, Dungannon and Omagh was not cost-effective with a mean ICER of £54,234/QALY under the old NICE threshold of £30,000. However, this project's mean of $£161,808$ is quite surprising considering it was expected the cost-effectiveness would increase, not decrease by over $3x$. This is likely due to extreme variation and instability of the ICER ratio and extremely low QALYs, or maybe the inclusion of the Dungannon, Cookstown and Omagh areas really increases the ICERs that much.

However, the present project's mean of £161,808 does not reflect a deterioration in cost-effectiveness relative to Cairns et al. (2011). Two factors explain the difference. First, Cairns et al. pooled Fermanagh with Cookstown, Dungannon and Omagh, whereas this project reports Fermanagh in isolation. Fermanagh is the largest and most sparsely populated of the 26 LGDs with both the lowest modelled OHCA incidence and the longest baseline EMS response times, so separating it from three less-extreme rural LGDs removes an averaging effect that lowered Cairns et al.'s estimate. Second, the mean of per-simulation ICERs is known to be unstable when incremental QALYs are small and right-skewed, as they are here (mean 2.6 QALYs/year), and is dominated by simulations in the upper tail. The $95\%CI$ (£26,191-£817,373), which spans more than an order of magnitude, reflects this.

Figures 4.8 and 4.9 shows that both allocation rules produced similar cost-effectiveness results, showing that $\approx 70\%$ of simulations were cost-effective under the old NICE WTP threshold of £30,000, and that $\approx 80\%$ of simulations were cost effective at the new under consideration NICE £35,000 WTP.

Figures A.5 and A.6 in the appendix demonstrate that the scheme is not cost-effective at all for the best NI-wide rules (B1 and B6), with less than 25% of the 1000 simulations ran lying within the £35,000 NICE WTP threshold for both implementations.

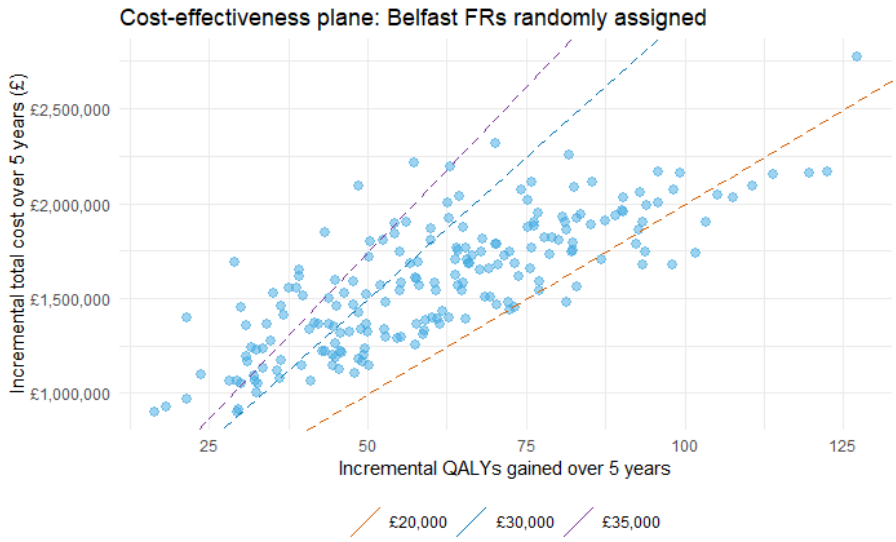


Figure 4.9: Cost-effectiveness plane (200 iterations) for Belfast under random FR allocation, showing incremental costs and QALYs over 5 simulated years (blue points). NICE willingness-to-pay thresholds of £20,000 (orange line), £30,000 (blue line), and £35,000 (purple line) per QALY are shown.

4.5 Sensitivity analysis

Figure 4.10 presents the one way sensitivity analysis for Carrickfergus which demonstrates that the schemes effectiveness (measured as the proportion of arrests saved) was most sensitive to the number of responders available per week and the probability that an individual FR attends a paged incident, each producing changes of approximately 1.5 percentage points of change. Dispatch sensitivity also had a significant influence, with the probability that a FR collected their AED producing more moderate effects.

Rota length had negligible impact on scheme performance, suggesting that the rota structure is not an important factor. This suggests that efforts to improve the schemes effectiveness are better directed toward volunteer recruitment and allocation strategies than rota design.

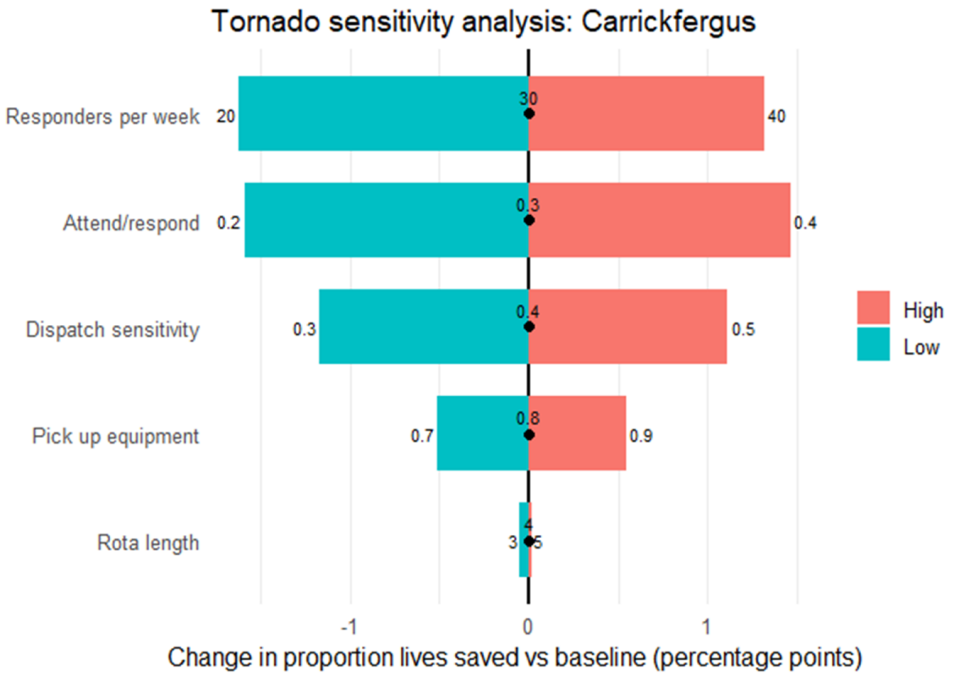


Figure 4.10: One way sensitivity analysis tornado diagram for Carrickfergus, showing the change in proportion of arrests saved by the FR scheme relative to the baseline scenario (30 responders per week, attend/respond probability = 0.3, dispatch sensitivity = 0.4, equipment pick-up probability = 0.8, rota length = 4 weeks). Each parameter is varied individually between its low and high value while all others are held at baseline. Bars extending to the right (red) represent the high scenario, while bars extending to the left (blue) represent the low scenario. The baseline value for each parameter is shown at the centre point.

Chapter 5

Discrete Event Simulation

5.1 Purpose

A report by the Northern Ireland Audit Office on ambulance handovers in NI [6] states that the recommended Northern Ireland Ambulance Service (NIAS) target handover time for a patient from an ambulance team to hospital staff is 15 minutes, yet only 7% of NIAS handovers reached this target [6] (figure 5.1) resulting in 25% of NIAS operational capacity wasted [6], and reaching an average of 93 minutes in Belfast in December 2024 [96].

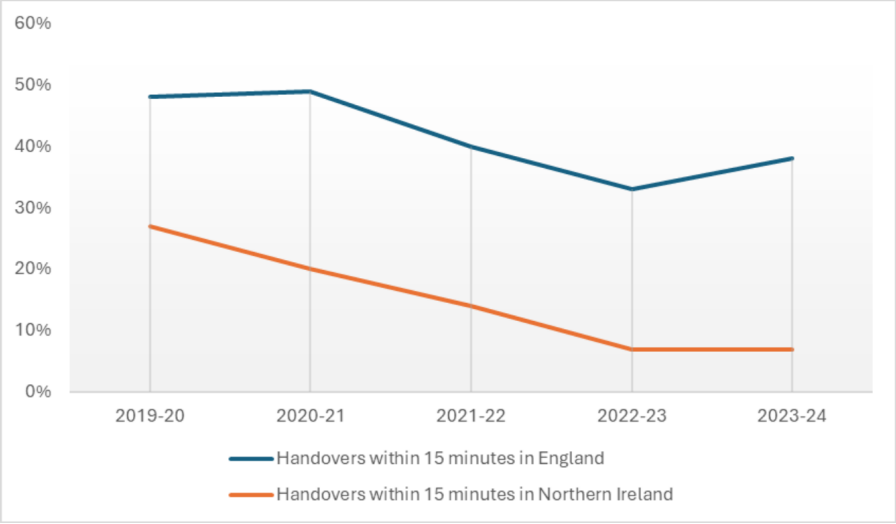


Figure 5.1: Percentage of Ambulance handover times to A&E within the 15 minute NHS target between 2019 and 2024 has rapidly fallen from 27% to 7% for NI (Orange). (Source: Figure 1 in the NI Audit Office report on ambulance handovers [6].

A discrete event simulation (DES) model was developed as a supporting component of this project to explore how time varying demand, priority queueing for ambulances, and

handover congestion to Accident & Emergency (A&E) interact to affect performance indicators such as queue wait times and ambulance response times. The purpose of this model is not to simulate NIAS in full detail, but to provide a simple yet defensible representation of a priority queueing model under pressure to examine assumptions and differing scenarios through one way sensitivity analysis.

5.2 Reason for choosing DES

DES was chosen as the problem is of queues, flows through states in a system such as scene time, transportation time etc, and constrained resources. Ambulance systems are characterised by time-varying arrivals, priority queues for calls, stochastic job durations and constraints such as handover times.

The choice of DES is also consistent with prior EMS and ambulance simulation literature. For example, Ingolfsson et al. (2003) developed a DES model to show that a single start (SS) station system for Edmonton EMS operations increased average ambulance availability and the fraction of calls reached within the department's response time standard. Lam et al. (2014) [56] utilised a DES model for Singapore EMS to evaluate the effectiveness of different dispatch policy alternatives, showing that modifications of the dispatch policy combined with the reallocation of existing ambulances reduced the gap between the ideal and existing response distribution by $11 - 13\%$.

5.3 Model scope

The DES of this project was set as a regional pooled ambulance model rather than a full spatial dispatch model. This was adopted because the main aim was to study queueing dynamics and the effect of handover delays and time-varying demand in an ambulance model under pressure, not to optimise station location or reproduce NIAS in full spatial detail. This is in line with Fonseca and Pearson's argument that DES is most useful to inform discussion of the underlying causalities and impacts of changes on the system instead of direct comparisons with reality [54].

An important family of models within DES is the hypercube queueing model. Hypercube models were developed to analyse spatially distributed emergency service systems and are useful when designing response districts for mobile ambulances, ambulance deployment or dispatch policies [97]. A more recent review by Iannoni and Morabito (2023) [98] shows that hypercube extensions are particularly useful for applications involving multiple dispatch, server reservation, cut off policies for high priority calls, queued call-priority rules and cross trained emergency resources.

A further useful extension is the incorporation of queued call priorities into the hypercube model. De Souza et al. (2015) [99] extended the hypercube model to analyse ambulance systems with multiple priority classes and a queue for emergency calls, and demonstrated its application in an urban EMS case study in Ribeirão Preto, Brazil. This study is important as it demonstrates that the hypercube model is useful for studying the effect of priority rules among waiting calls to evaluate important performance measures such as mean response time. This links closely to the present study where clinical priority is also central, however the current supporting model was preferred because the objective

was to study queueing pressure, conveyance (the transportation of ambulance services to hospital) and handover delay in a simple more flexible DES model.

Papakonstantinou (2025) [100] examined over 850,000 London Ambulance Service emergency (LAS) calls from 2023 to assess how Call category priorities predicts transport and patient disposition. The study found a clear link between call acuity and emergency transport, showing emergency transport rates of 74.1% for Category 1 and 72.7% for Category 2, compared with 5.9% for Category 3 and 1.2% for Category 4. Lower priority calls more often did not require immediate transportation to hospital, with approximately 40% of Category 3 calls resolved on scene or referred to alternative pathways. The paper also highlighted that call categorisation and final transport decisions do not align perfectly, with some lower priority calls escalating to urgent transport and some higher priority calls ending in non-conveyance. These findings are useful for the current DES because it provides recent empirical statistics for category specific conveyance behaviour while also acknowledging discrepancies between initial dispatch categories and patient condition.

5.4 Data inputs

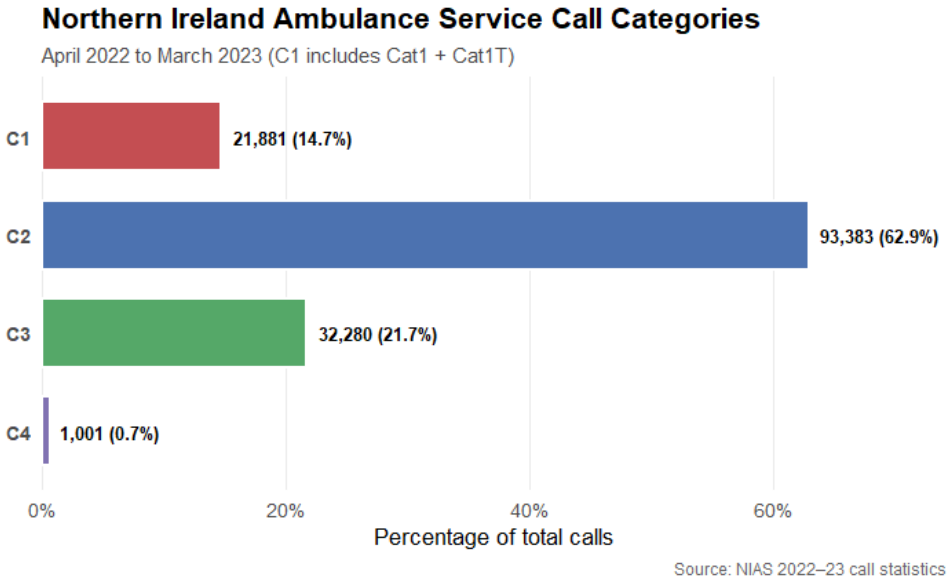


Figure 5.2: Number of NIAS calls received by category produced using April 2022 to March 2023 NIAS data [101]. Category 1 (C1) urgent life threatening calls is in red, Category 2 (C2) serious but not immediately life threatening calls is in blue, Category 3 (C3) urgent but non life threatening calls is in green, Category 4 (C4) non urgent calls is in purple.

The annual attended incident total [5] is converted to a weekly demand profile across 168 cells representing each hour using South East coast demand profiles [66], shown in figure A.7. Category specific weekly demand is obtained by multiplying the weekly demand by each category's shares according to an NIAS 2022/23 Freedom of Information request [101].

Input	Source	Value used
Annual attended incidents	NIAS 2024/25 statistics [5]	173,695 incidents/year
Call category mix (C1/C2/C3/C4)	NIAS 2022/23 FOI [101]	14.7% / 62.9% / 21.7% / 0.7%
Time-of-day demand profile	NHS England 2024/25 [66]	7×24 matrix of attendance shares
Ambulance staffing (DCA)	NI Assembly answer 2025 [63]	60-65 day shift, 47-52 night shift: 80% dispatchable (Assumption)
Category conveyance probabilities	London Ambulance Service FOI, Papakonstantinou et al. [100]	0.741 / 0.727 / 0.059 / 0.012
Clinical handover time	NHS target [5] and Belfast observation [96]	calibrated to NHS 15-min target; total handover calibrated against Belfast 93-min mean
Maximum queue wait before renege	Clinical judgement (Assumptions)	C1/C2: no cap; C3: 12 hours; C4: 6 hours

Table 5.1: Data inputs for the DES.

5.4.1 Call arrivals

Calls arrive by a non-homogenous Poisson process. For day $d \in \{1, \ldots, 7\}$ and hour $h \in \{0, \ldots, 23\}$ the category specific arrival rate is given as

$$\lambda_c(d, h) = W_c \, P(d, h), \tag{5.1}$$

where W_c is the weekly call total for category c and $P(d, h)$ is the normalised demand profile form NHS England data [66], with $\sum_{d,h} P(d, h) = 1$. Inter-arrival times within each hour block are drawn from an exponential distribution with $\lambda_c(d, h)/60$, converting hourly rates to per minute rates.

5.4.2 Ambulance queue and Service-time distributions

Calls then compete for a pool of ambulances whose capacity varies throughout the week according to the NIAS schedule [63] (see Section 5.4). A priority queue then serves higher priority calls first (C1 → C2 → C3 → C4), with ties within categories being first come first serve. Once an ambulance is allocated the call proceeds through the job cycle shown in Table 5.2.

Every job cycle component is modelled as a Gamma distribution with mean μ and shape parameter α. For a $\Gamma(\alpha, \theta)$ distribution the mean is $\alpha\theta$ and variance $\alpha\theta^2$, so setting the scale parameter as $\theta = \mu/\alpha$ produces a distribution with the desired mean while letting α control how tightly values vary around it. Smaller α gives a wider more right skewed distribution, while larger α gives a tighter and more symmetric one.

Table 5.2 lists the means μ and shape parameters α used. These values combine published UK ambulance benchmarks with modelling assumptions:

1. **Mobilisation** (1.5 minutes) is a modelling assumption. UK ambulance services don't publish mobilisation times, but usually state that it is quite low [102].

Component (minutes)	**C1**	**C2**	**C3**	**C4**	**Shape α**
Mobilisation	1.5	1.5	1.5	1.5	10
Transport to A&E	7.0	16.0	20.0	22.0	4
Scene time	35.0	28.0	22.0	18.0	6
Transport to A&E	7.0	16.0	20.0	22.0	6
Clinical handover	17.0	15.0	13.0	12.0	6
Clearance	6.0	6.0	6.0	6.0	10
Conveyance probability	0.741	0.727	0.059	0.012	–

Table 5.2: Job cycle service-time means and gamma shape parameters. Each component is sampled from a $\Gamma(\alpha, \mu/\alpha)$ distribution where μ is the category-specific mean and α is the shape parameter shown in the final column.

2. **Travel to scene** is calibrated for the higher priority calls. The C1 and C2 means are set so that mobilisation plus travel matches the corresponding NIAS mean response-time targets of 8 and 18 minutes respectively [103]. However, lower priority calls values are mostly modelling assumptions, as NIAS publishes only 90th percentile targets for the C3 and C4 targets.

3. **Scene time** (35 minutes for C1 down to 18 minutes for C4) is a modelling assumption. The C1 value is informed by the cardiac arrest literature, where a Vienna Cardiac Arrest Registry study found on scnee times under 35 associated with higher survival for OHCA, and a US EMS system mandated a 30 minute on scene interval for OHCA resucitation [104]. While UK ambulance services do not publish scene times by priority category, the decrease from C1 to C4 reflects the operational difference between prolonged on scene intervention (such as CPR) for high priority calls and brief "see and treat" assessments for low priority calls.

4. **Transport to A&E** mirrors travel to scene time exactly (7-22 minutes across all categories).

   Clinical handover (12-17 minutes centred on the 15 minute NHS target [6]) represents the time it takes for the ambulance crew to hand the patient over to A&E staff once a bay is free and passing on information about their condition. It's separate from the time the ambulance spends queueing for a bay in the first place, which is modelled by another queuing model (Section 5.4.3). The NHS target is 15 minutes [6], but observed total handover times are much longer at 31 minutes nationally in July 2024 [105] and 93 minutes in Belfast December 2024 [96] as most of that time is spent queueing for a bay.

5. **Clearance** (6 min) is a modelling assumption representing vehicle restock and reset.

6. **Conveyance probability** is data-driven. Values taken directly from an FOI-based analysis of over 850,000 London Ambulance Service calls in 2023 [100]. High-acuity categories are almost always conveyed (C1 74%, C2 73%), low-acuity categories rarely are (C3 6%, C4 1%).

5.4.3 Handover queueing

When an ambulance arrives at A&E it receives a handover slot from a fixed pool. If all slots are occupied, the ambulance joins a first come first serve queue. Once a slot becomes free the clinical handover for the next ambulance begins, where the ambulance remains for the entire wait and clinical handover period so the total handover duration is

$$T_{\text{handover}} = T_{\text{queue for slot}} + T_{\text{clinical}}. \tag{5.2}$$

The handover queue is modelled without call category based priority as once an ambulance is at A&E, the original call category is no longer the primary triage indicator. A&E staff do an initial assessment based on clinical condition and how stable the patient is based on arrival [106], so a priority queueing model similar to calls would be less realistic.

Northern Ireland has nine Type 1 A&E departments serving NIAS arrivals. Rather than model each hospital as its own queue (which would require hospital specific arrival rates and handover slots not published in public data), the model pools regional handover capacity into a single resource of c simultaneous slots. The baseline $c = 5$ is a modelling choice as it represents about half the number of NI Type 1 EDs, and produces a utilisation $\rho \approx 0.5$ (percentage of time that a server is busy) which is stable and allows the model to handle the high demand during nightly periods without the backlog never being dealt with (continuously rising throughout the simulation). A configuration of $c = 4$ is saved for the sensitivity analysis in Section 5.5.3, where it pushes utilisation to saturation ($\rho \to 1.00$) and reproduces conditions similar to the 93 minute Belfast handover mean observed in December 2024 [96].

Modelling handover in this way is important because the 93 minute figure is not the result of a slow clinical procedure but of ambulances waiting for hospital capacity while A&E is under pressure.

5.5 Results

The simulation was implemented in R using `simmer` package and ran for 15 replications over the course of 60 simulated days each, with a 14 day burn in period to stabilise queues.

5.5.1 Baseline scenario

Figure 5.3 shows the ambulance fleet alongside demand across the week. Demand oscillates between approximately 10 incident per hour overnight and 30 per hour in the late afternoon, while sustainable throughput ranges from 35 to 45 calls per hour depending on the shift. The demand remains under ambulance fleet at all times, indicating that under this model the ambulance fleet has sufficient capacity to handle incoming calls at an average rate.

The average utilisation of the ambulance fleet over the simulation was approximately 55% with peaks of around $80\% - 90\%$ during evening hours. Handover capacity utilisation was approximately 55% with evening peaks of around 70%.

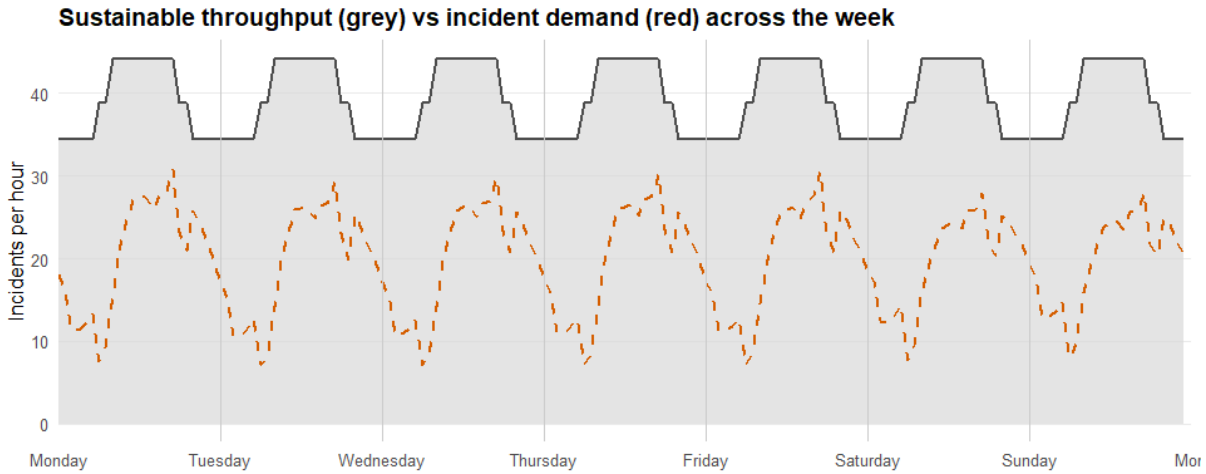


Figure 5.3: Ambulance fleet size throughout the week (grey) plotted against incident demand (orange) throughout each day of the week, generated using R.

5.5.2 Queue contribution to response time

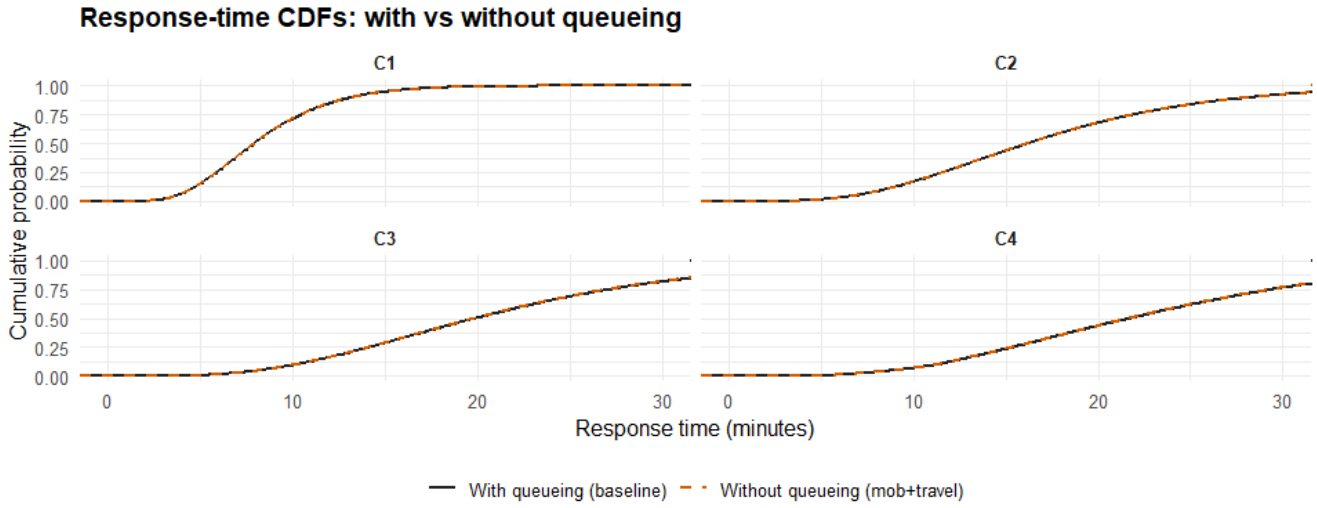


Figure 5.4: Cumulative distribution of response times by category, with (solid black line) and without (dashed orange line) queueing. Curves are visually identical at all categories, where the call queue contributes negligibly to response time under this model.

Figure 5.4 shows that there is a negligible contribution by call queues to response time across all categories. This result is consistent with empirical data, where in the February 2026 NHS England Ambulance Quality indicator statistical note the mean 999 call answer time was 3 seconds [107].

Figure 5.5 shows that this differs for handover queues, where the two curves separate by approximately 1.8 minutes at the median and up to 4.6 minutes at the 90th percentile. The simulated mean total handover duration was 17.8 minutes (95% CI 17.66 – 17.94), exceeding the 15 minute NHS handover target [6] despite the relatively optimistic model assumptions for the baseline.

Figure 5.6 shows that the mean handover slot queue length varies throughout the day. The queue is close to zero overnight, begins building in late morning, and peaks at an average

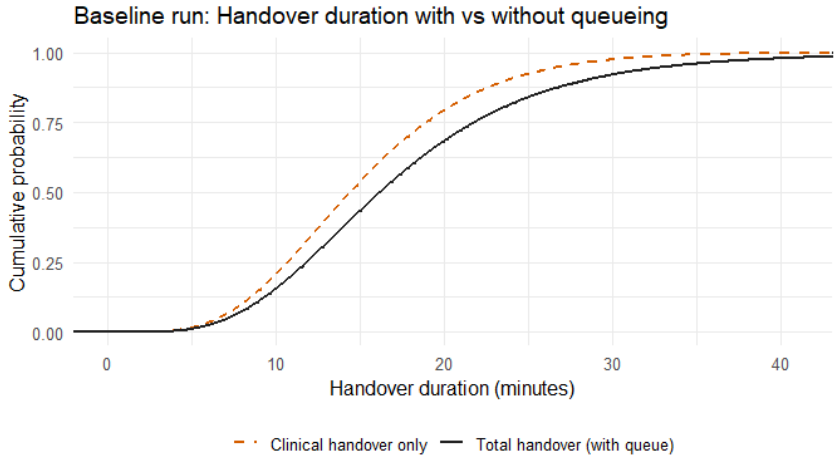


Figure 5.5: Cumulative distribution of handover duration for the baseline scenario: total handover (solid black) versus clinical handover alone (dashed orange). The gap between curves is the contribution of waiting for a handover slot

of approximately 1.3 at 18:00 before draining into the evening. Figure A.9 confirms that this queue drains consistently overnight across all replications, with no constant backlog across days. These patterns are also shown for the mean call queue length in figure A.12 in the appendix.

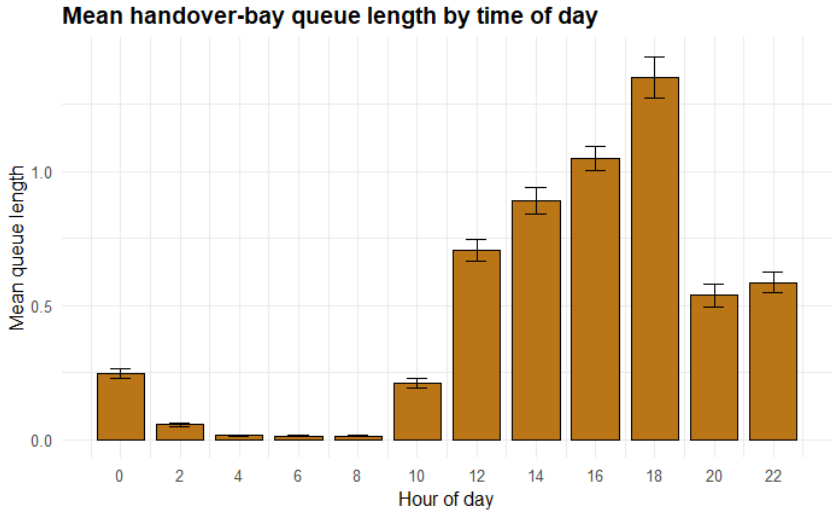


Figure 5.6: Mean handover-bay queue length by time of day. Error bars show 95% confidence intervals across 15 replications

5.5.3 Sensitivity analysis

One way sensitivity analysis was tested on incident demand, ambulance staffing, handover capacity and conveyance probability (probability that the patient is brought back to hospital from the scene). Demand, ambulance staffing and conveyance were varied by 10% around the baseline, while handover capacity varied by ± 1 around $c = 5$ bays. Each scenario used 10 replications of 60 simulated days, with a 14 day burn in period of each replication.

5.5.3.1 Call queueing times

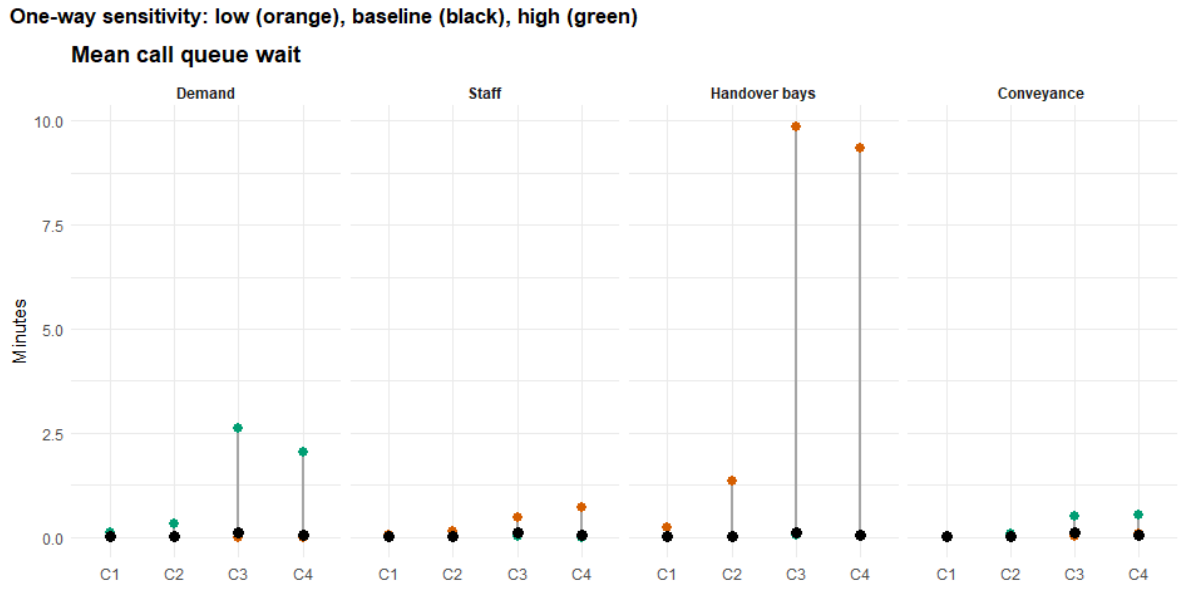


Figure 5.7: One-way sensitivity analysis of the affects of varying demand (number of calls), staff, and conveyance probability by 10% on the mean call waiting time across 10 simulation and 60 days. Handover bays are also varied by ±1. High (+10% on baseline) is shown in blue while low (−10% on baseline) is shown in orange.

Figure 5.7 shows the mean call queue waiting time by category for each sensitivity axes. It shows that C1 and C2 call waiting times are almost invariant to every varied parameter, reflecting their high priority in the call queueing system. C3 and C4 calls absorb most of the effects due to their lower priority, which is to be expected.

It also demonstrates that handover capacity is by far the most dominant parameter, where removing one handover slot drives C3 and C4 call waiting times to approximately 10 minutes. Demand is shown to have a more moderate affect, where decreasing the demand by 10% increases the means of the lower priority calls to 2.5 minutes. Decreasing staffing by 10% (rounded) and increasing conveyance by 10% is shown to have a negligible impact compared to demand and handover capacity.

5.5.3.2 Handover queue waiting time

Figure 5.8 shows the one-way sensitivity analysis with the same four parameter changes applied to the mean handover queue waiting time across all categories and 10 simulation runs. Removing one handover bay increased the mean handover wait time from 2.5 minutes to 13.6 minutes, whereas adding one bay reduced it to 0.7 minutes. This demonstrates that loss of handover capacity has a much greater effect than gaining handover capacity, implying that even small reductions in available A&E handover capacity produces incredibly large effects on ambulance waiting times at A&E which could be used for responding to other calls. Demand and conveyance produces moderate and symmetric effects, while staffing had no effect on handover queues as expected since ambulance staffing do not affect A&E capacity.

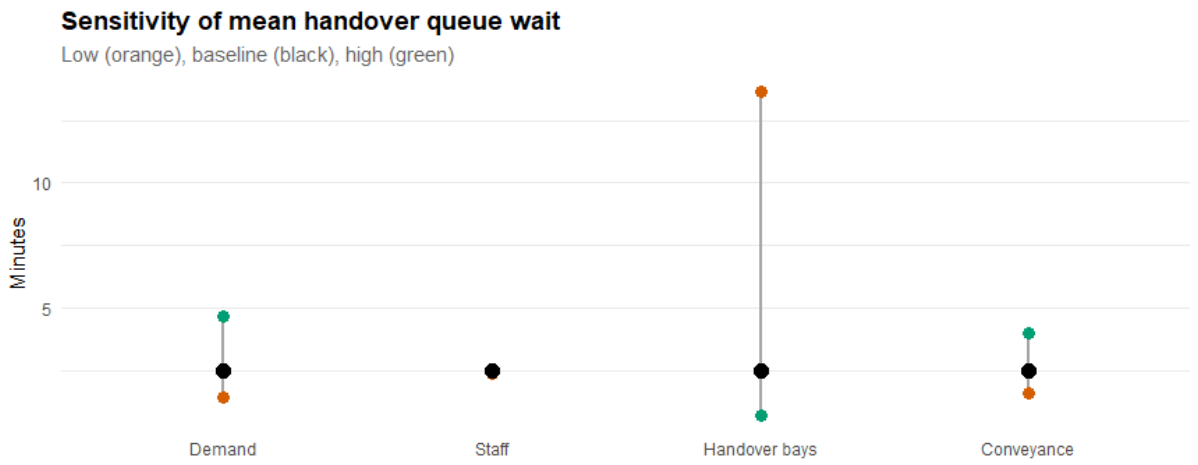


Figure 5.8: One-way sensitivity analysis of the affects of varying demand (number of calls), staff, and conveyance probability by 10% on the mean handover waiting time across 10 simulation and 60 days. Handover bays are also varied by ±1. High (+10% on baseline) is shown in blue while low (−10% on baseline) is shown in orange, and the baseline is shown in black

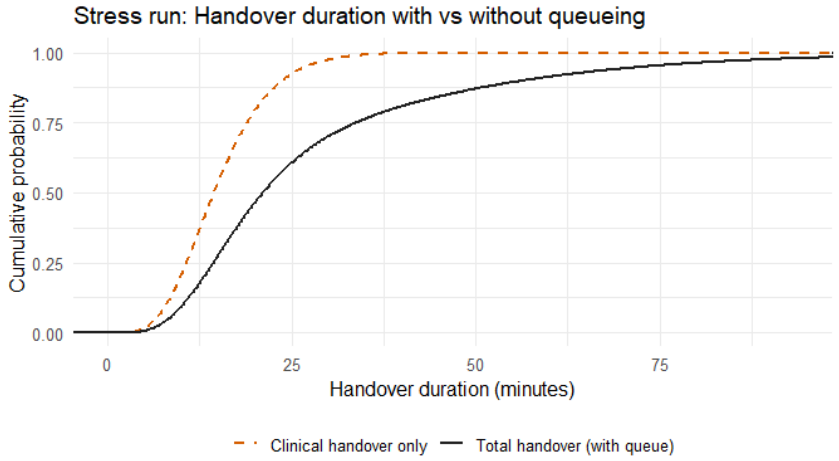


Figure 5.9: Cumulative distribution of handover duration for the stress scenario (one less handover slot): total handover (solid black) versus clinical handover alone (dashed orange). The gap between curves is the contribution of waiting for a handover slot

5.5.3.3 Stress scenario: handover capacity reduced by one bay

Figure 5.9 compares the distribution of total handover duration against the clinical handover component alone for a stress scenario of reducing the handover bay spaces from 5 to 4 slots, ran over 50 simulations across 60 days with a 14 day burn in period.

Under this stress scenario the two cumulative distribution curves separate dramatically, with the median total handover duration (wait for handover bay + clinical handover) rising from 16.2 minutes for no queueing to 21.1 minutes with queueing. The tail widens more severely with the 90th percentile of total handover duration moving from 28.4 minutes to 56.5 minutes, and the 95th percentile from 33.4 minutes to 73.0 minutes. This widening is not caused by any change in the clinical handover process but entirely by the queueing resulting from the decrease in handover bays.

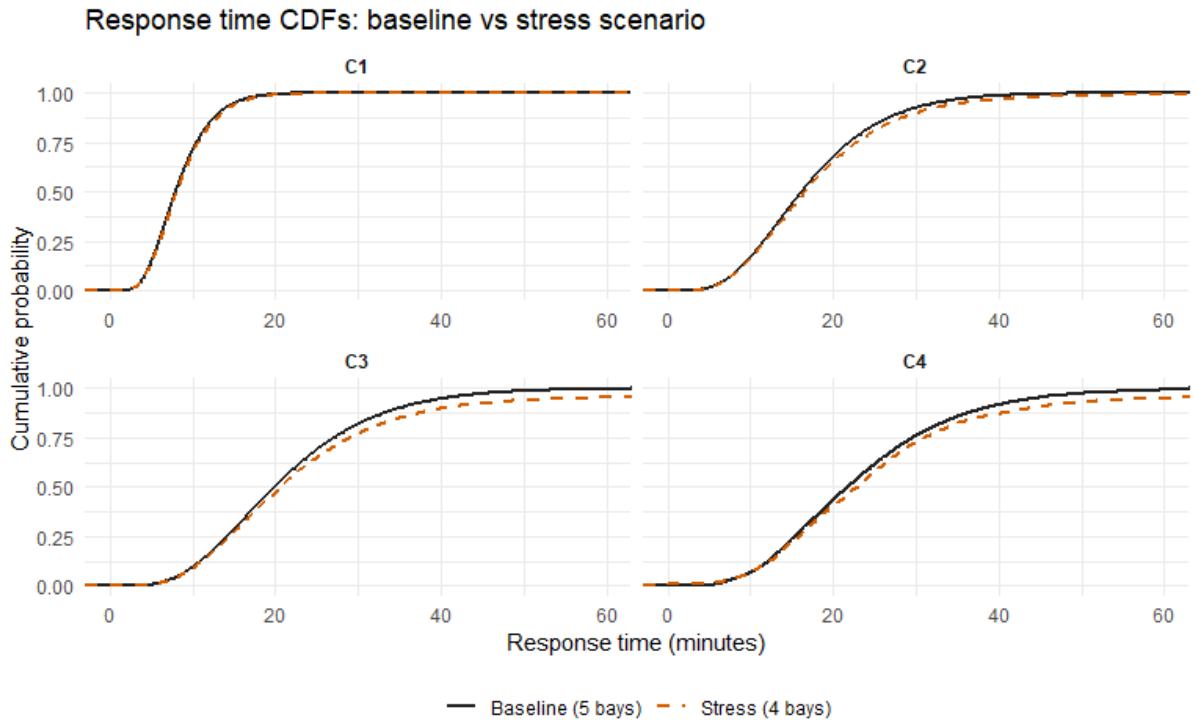


Figure 5.10: Cumulative distribution of response times by category, comparing the baseline (5 bays, solid black) against the stress scenario (4 bays, dashed orange).

Figure 5.10 shows the effect of removing one handover bay on the overall response times of the model, where C1 and C2 response times are essentially unchanged while C3 and C4 response times show a visible increase in response times (approximately 3 minutes at the median). This shift is due to the increase in ambulances being held waiting at A&E who are then unable to respond to calls, inflating call queues which then affect the response times for lower priority calls.

Chapter 6

Conclusion

This project aimed to demonstrate the usefulness of two simulation based techniques that are implemented widely in healthcare decision making, applying Monte Carlo simulation to assess the cost-effectiveness of a mobile volunteer FR scheme across Northern Ireland and Discrete Event Simulation to examine queueing dynamics in the NI Ambulance Service. Together, these findings demonstrate the value of simulation based modelling in supporting healthcare decisions that cannot easily be informed by trial data alone.

The Monte Carlo simulation model was used to assess the cost-effectiveness of a mobile volunteer First Responder scheme across Northern Ireland. The purpose of this model was to apply up to date 2021 NI census data, updated Life-tables, revised long-term neurological impairment survival evidence from Dillenbeck et al. (2024) [11], and a new proposed NICE willingness to pay threshold of £35,000 to the work done by Cairns et al. (2011).

The model used a Poisson process to generate ventricular fibrillation (VF) out-of-hospital cardiac arrests (OHCAs) using 2021 NI census data across all its Data Zones, log-logistic accelerated failure time model (AFT) to simulate ambulance response times (parametrised by Cairns et al. (2011) [1]), the Valenzuela et al. (1997) logistic regression model to simulate survival using these response times, and a Markov microsimulation incorporating NI life tables and Dillenbeck et al. (2024) [11] hazard ratios for long-term survival across three neurological impairment states. First responders (FRs) were then added with their response time also being simulated by a log-logistic AFT model parametrised by Cairns et al. (2011, and then compared with the EMS only scenario to feed into the Markov microsimulation for long term outcomes like Quality Adjusted Life Years (QALYs) and Incremental Cost Effectiveness Ratios (ICERs).

Varying ways of allocating FRs (allocation rules) were then set and the model ran for each to find that distributing FRs weighted by the expected number of OHCAs (incidence allocation) was the best rule, as expected. It was also found that NI wide incidence allocation outperformed within-LGD (Local Government District) allocation by allowing resources to be concentrated in the highest incidence urban LGDs (primarily Belfast) rather than

evenly distributed to more rural LGDs. Area adjusted allocation rules underperformed (even worse than randomly), which further highlights the importance of allocating FRs toward demand.

Cost-Effectiveness Analysis (CEA) was then performed across three contrasting LGDs using the within-LGD incidence allocation rule to show how demographic and geographic differences can change the effectiveness of the FR scheme, with Fermanagh performing poorly with a mean ICER of £161,808 per QALY (95% CI £26,191–£817,373) and Belfast performing well with a mean ICER of £26,509 per QALY (95% CI £18,998–£37,637). In fact, this combination of LGD and rule was the best performing one, even falling under NICE's old £30,000/QALY threshold suggesting that the scheme is moderately cost-effective with around 70% of simulations falling under the threshold. These are the sort of results that allows policy makers from NICE to make informed decisions and quantify risk on a particular scheme or medicine.

The DES model developed in Chapter 5 was a supporting component of this project, and its findings have implications for the main FR scheme conclusion. The DES showed that call queueing contributes negligibly to response time under baseline conditions, but that handover capacity at A&E has a dominant effect on the system. Removing one handover bay increased the mean handover wait from 2.5 minutes to 13.6 minutes, with lower priority call response times also rising by approximately 3 minutes at the median. This indicates that current pressure on NI ambulance response times is affected heavily by handover delays rather than by a shortage of ambulances, consistent with the NI Audit Office report (2025) [6] and the 93 minute mean handover observed in Belfast in December 2024 [96].

Further work includes investigating how inflation on the inputted costs from the Department of Health's 2006 figures changes the cost-effectiveness analysis in this project, where it is estimated by the Bank of England to be at +75.1% from 2006-2026 [108]. In hindsight, implementing the AVID ratios [58] exactly like Cairns et al. (2011) would have brought proper insight into how the model has changed with the implementation of new data without worrying about changes to their method. Another further works extension would be is combining the DES aspect in Chapter 5 with the first responder scheme studied to reflect the increasing handover delays, such as the 93 minute average handover time observed by the NIAS in Belfast in December 2024 [96]. This would require access to updated and more accurate ambulance response times and job cycle component information than the assumptions of their distributions presented in Chapter 5.

In summary, this report has identified the benefit of implementing a well managed FR scheme in more urban areas of Northern Ireland, such as Belfast under a within-LGD incidence allocation rule (A1). However, the scheme is unlikely to be cost-effective in the most rural LGDs such as Fermanagh, reflecting the difficulty of achieving meaningful FR coverage in sparsely populated areas even with a 15-mile paging radius. This highlights a tension between efficiency and equity in FR allocation, since strategies that maximise overall QALY gain concentrate resources in urban LGDs and leave rural populations with the longest EMS response times receiving the fewest additional responders. NICE's 2025 update to its Health Technology Evaluations manual formally introduced distributional cost-effectiveness analysis (DCEA) as a way of addressing such equity considerations in Health Technology Assessment [109], and applying DCEA to this project's model would be a natural extension of this work.

Appendices

Appendix A

Supplementary Figures

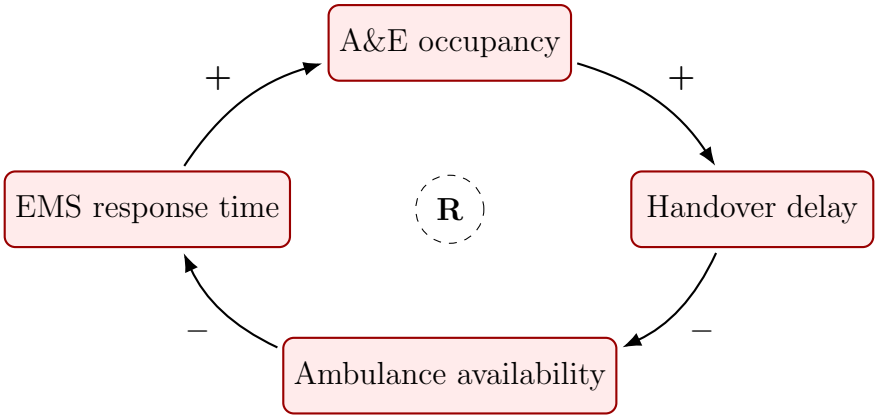


Figure A.1: An illustrative causal loop diagram of the reinforcing feedback loop between A&E occupancy, handover delay, ambulance availability and EMS response time. Arrow polarities show whether a change in the source variable increases (+) or decreases (−) the target variable. The two negative links combine to a reinforcing loop (R). Diagram conventions follow Sterman (2000), application to the NIASA&E system is original, and is motivated by the patient-flow study of Lane, Monefeldt and Rosenhead (2000).

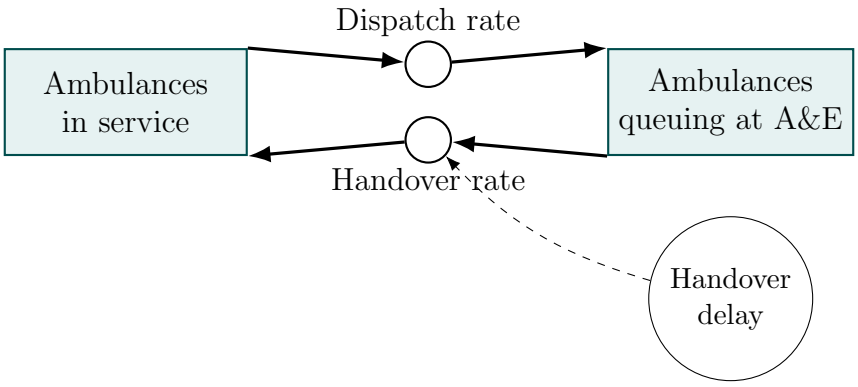


Figure A.2: Illustrative stock-and-flow representation of the ambulance handover system implied by the causal loop diagram in Figure A.1. Rectangles are stocks (ambulances in each location), thick arrows with valves are flows (rates at which ambulances transfer between stocks), the circle is an auxiliary variable influencing a flow rate via link (dashed arrow). Diagram conventions follow Sterman (2000).

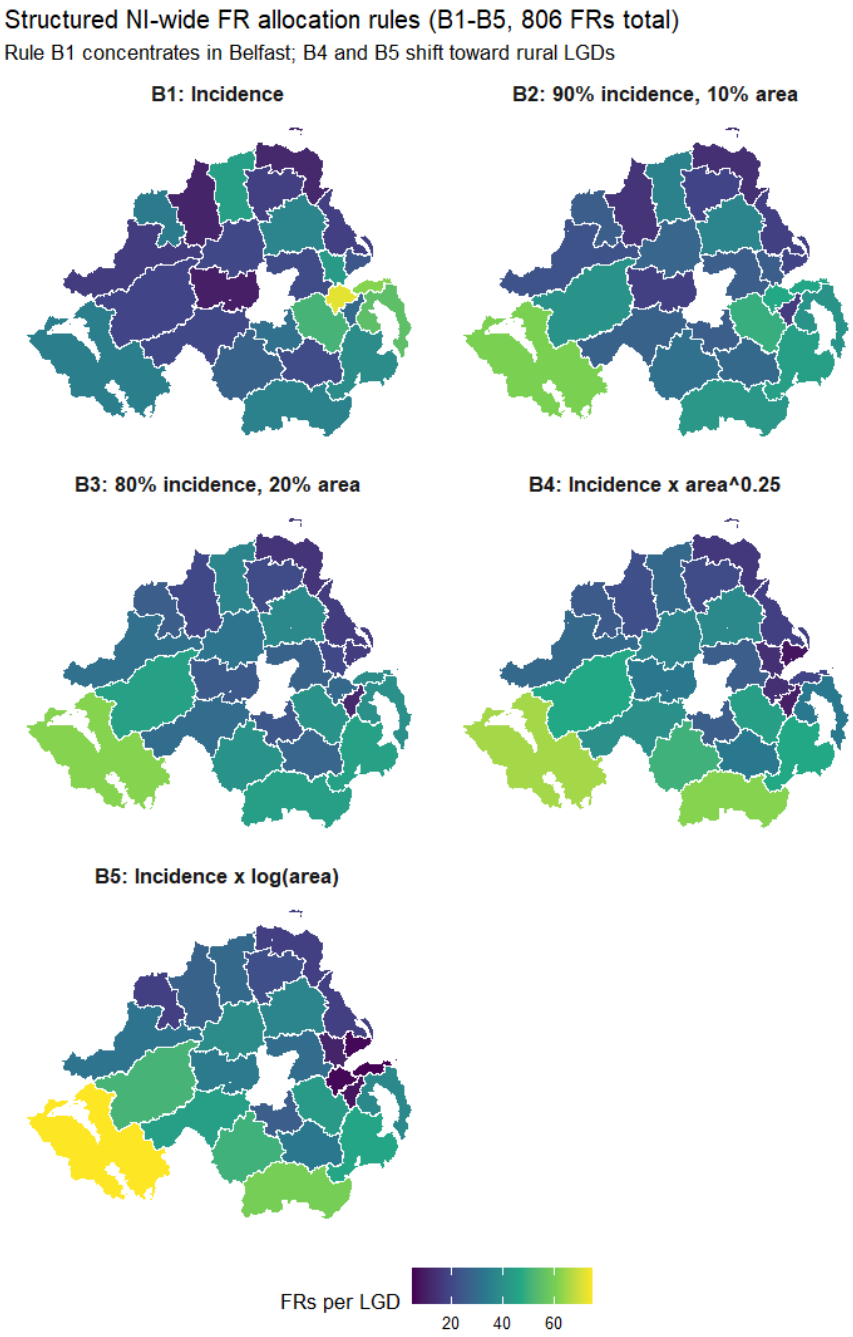


Figure A.3: Distribution of FRs by LGD under the five structured NI-wide allocation rules B1-B5 (Table 3.1), each distributing 806 FRs according to the rule's scoring function. Rule B1 (pure incidence) concentrates responders heavily in Belfast and other urban LGDs. Rules B2 and B3 soften this by adding a 10% and 20% area weighting respectively, shifting responders toward larger rural LGDs. Rules B4 and B5 apply a multiplicative area adjustment, producing progressively greater allocation to Fermanagh and other sparsely populated regions

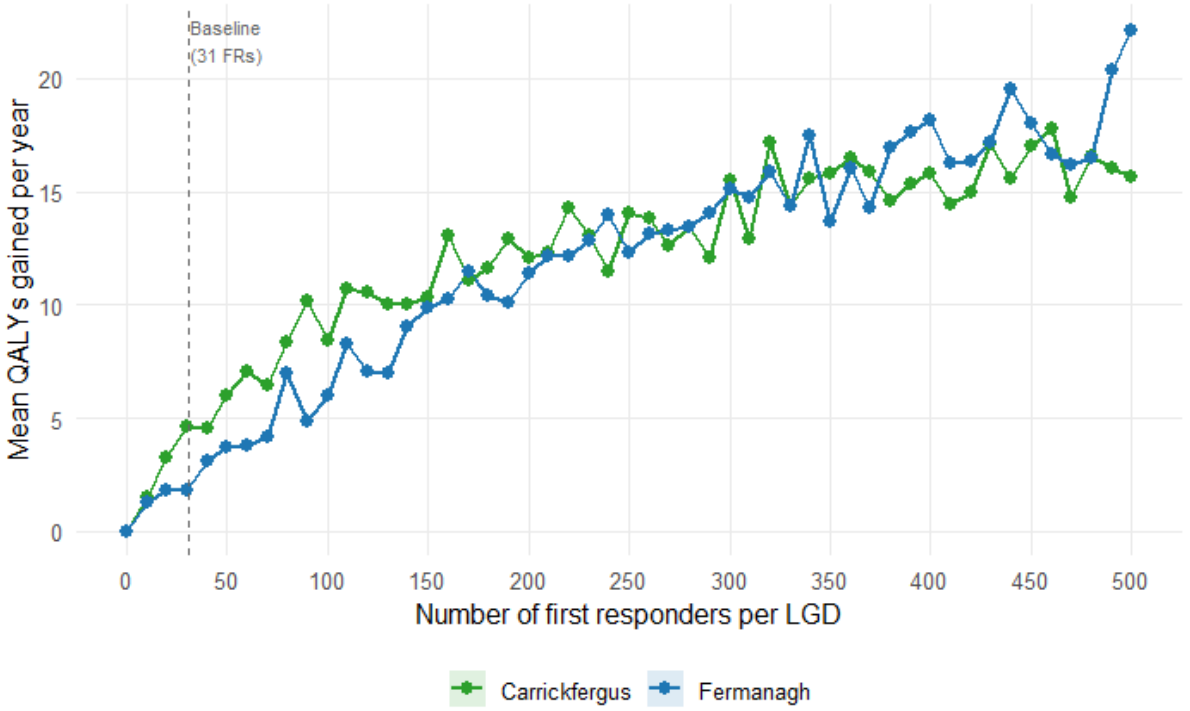


Figure A.4: QALYs gained with FRs ran using 20 simulations for each FR number, going up in increments of 10.

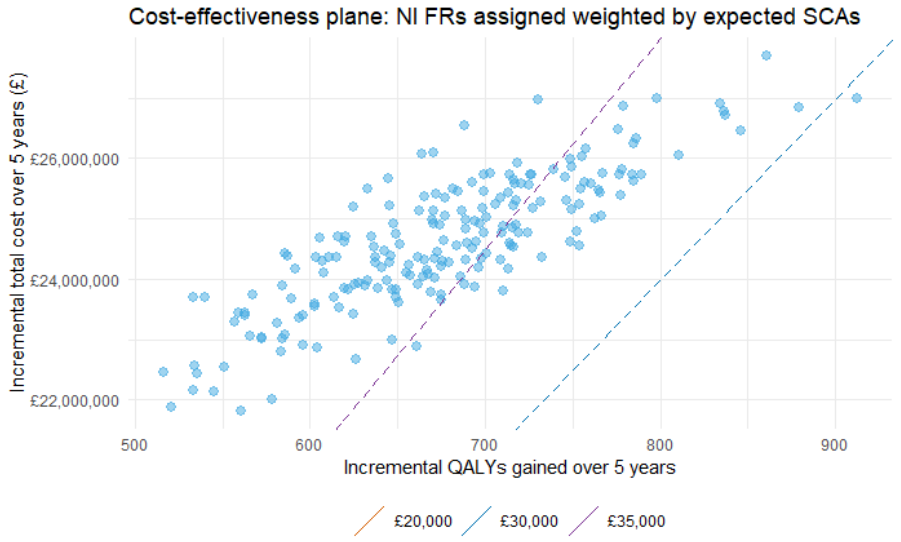


Figure A.5: Cost-effectiveness plane (200 iterations) for all of NI by expected incidence (rule B1), showing incremental costs and QALYs over 5 simulated years (blue points). NICE willingness-to-pay thresholds of £20,000 (orange line), £30,000 (blue line), and £35,000 (purple line) per QALY are shown.

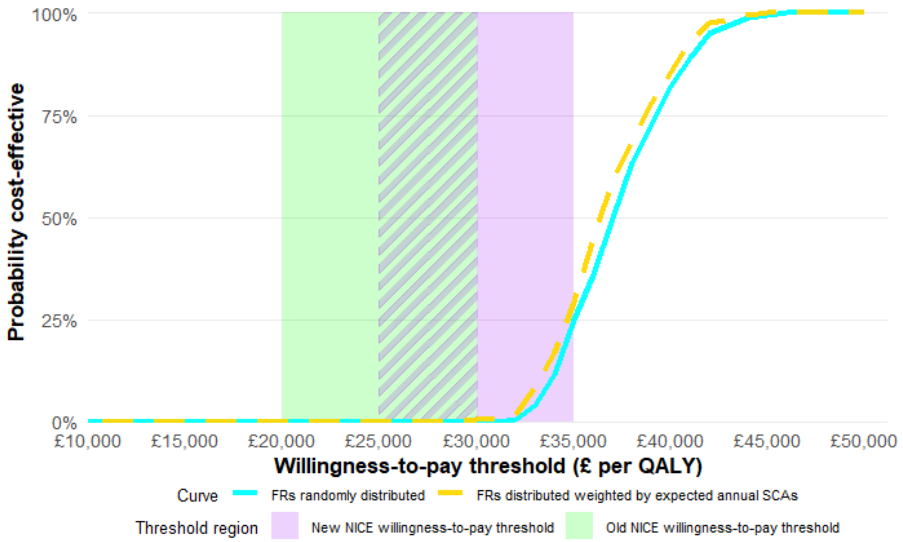


Figure A.6: Cost-effectiveness acceptability curves (CEACs) for the FR scheme being implemented to all of NI with FRs distributed to Data Zones randomly (cyan, rule B6 in Table 3.1) and FRs distributed weighted by expected number of annual SCAs for each Data Zone (gold, rule B1). These curves are shown against previous (green) and updated (purple) NICE willingness-to-pay thresholds, £20k-£30k and £25k-£35k respectively.

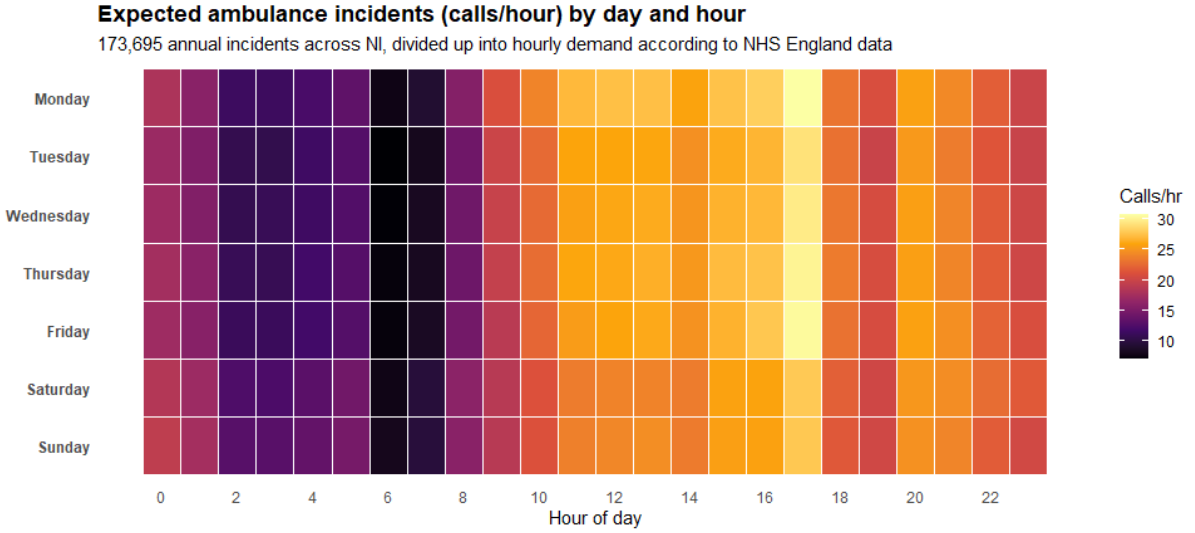


Figure A.7: Expected ambulance incidents per hour across the 7×24 weekly lattice generated for this project in R using NHS demand profiles [66] and NIAS call data [5][101]. Category shares are fixed so the same pattern applies across all four categories, with only totals differing.

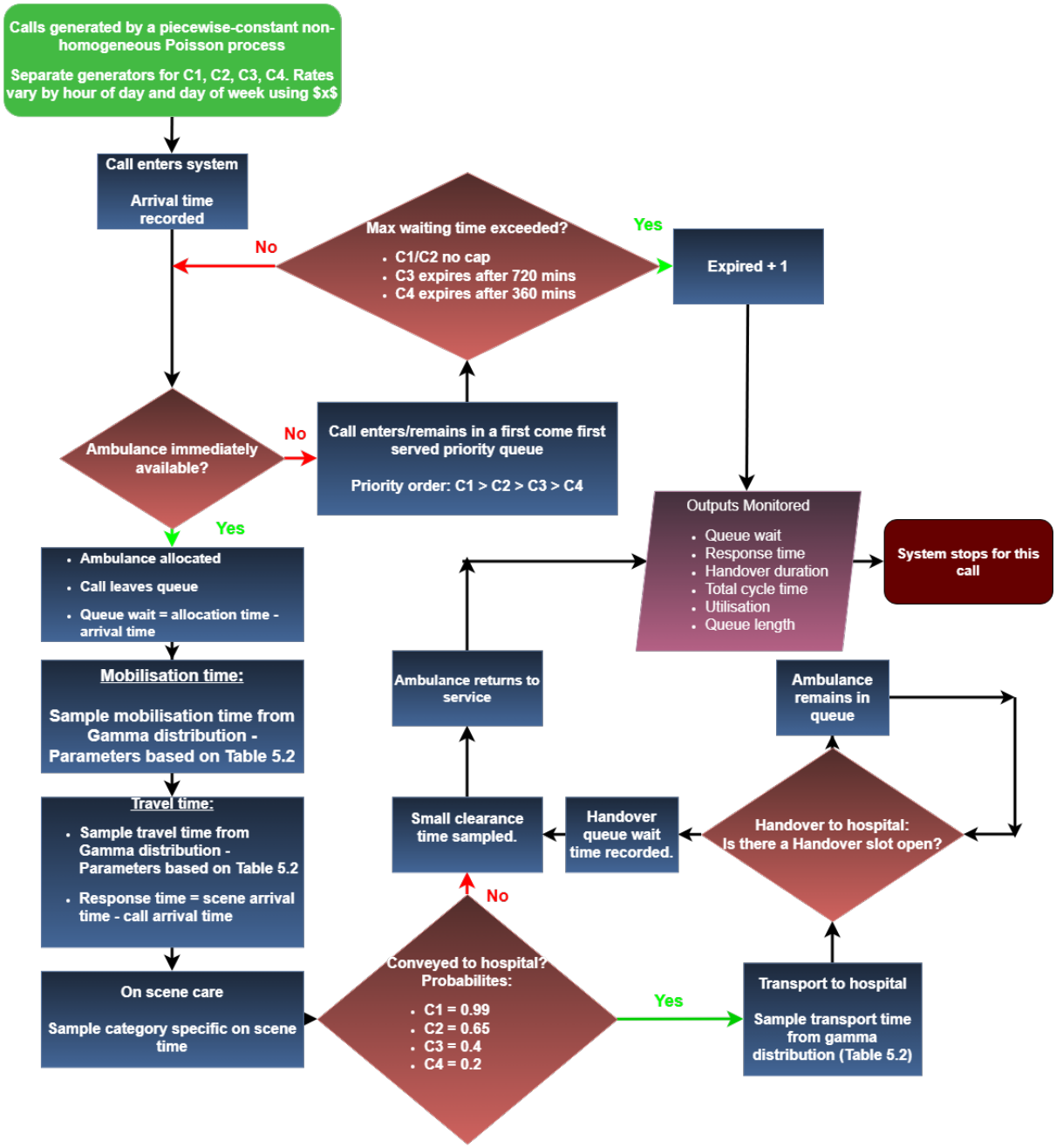


Figure A.8: A flowchart of how the DES ambulance model works in the R package *Simmer*. The blue squares represents processes, the red triangles represents decisions, and the purple parallelogram represents the recording of data for the model results.

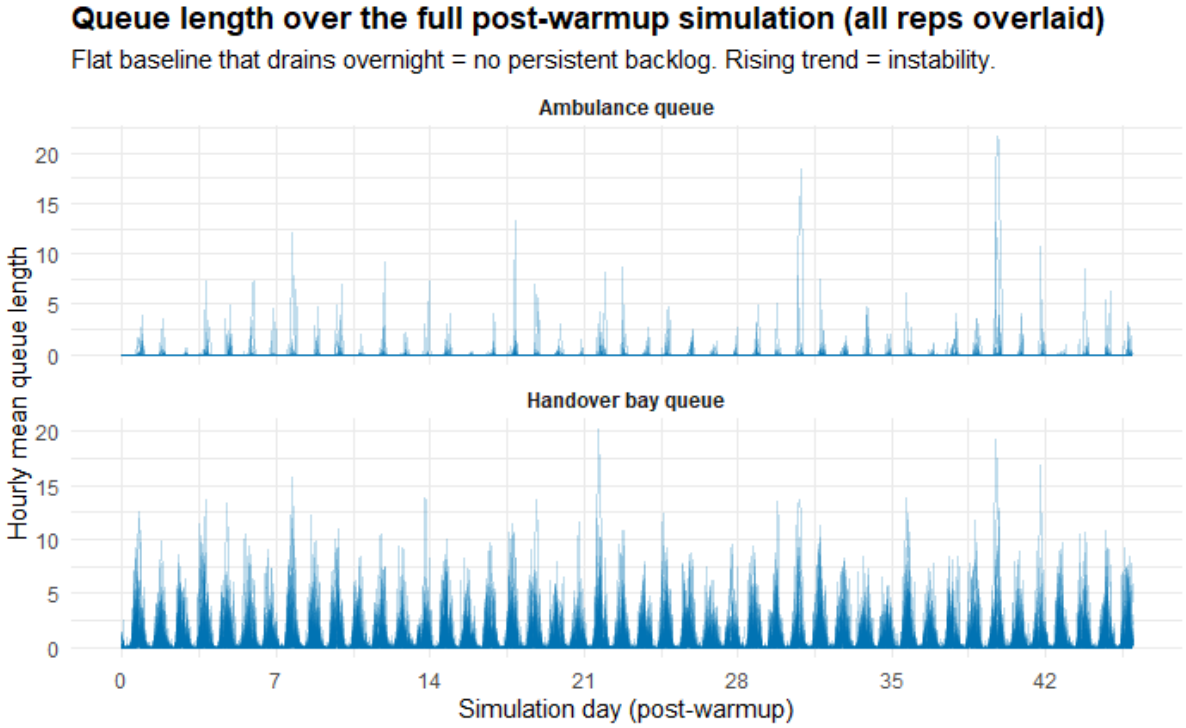


Figure A.9: Hourly time-weighted mean queue length across the full post-warm-up period (45 simulated days), with all 15 replications overlaid. Top panel: ambulance queue (calls waiting for an available ambulance). Bottom panel: handover-bay queue (ambulances waiting for a free AE handover slot). Both queues return to near-zero each night and do not trend upwards across days, indicating that the baseline scenario reaches a stable steady state with no persistent backlog.

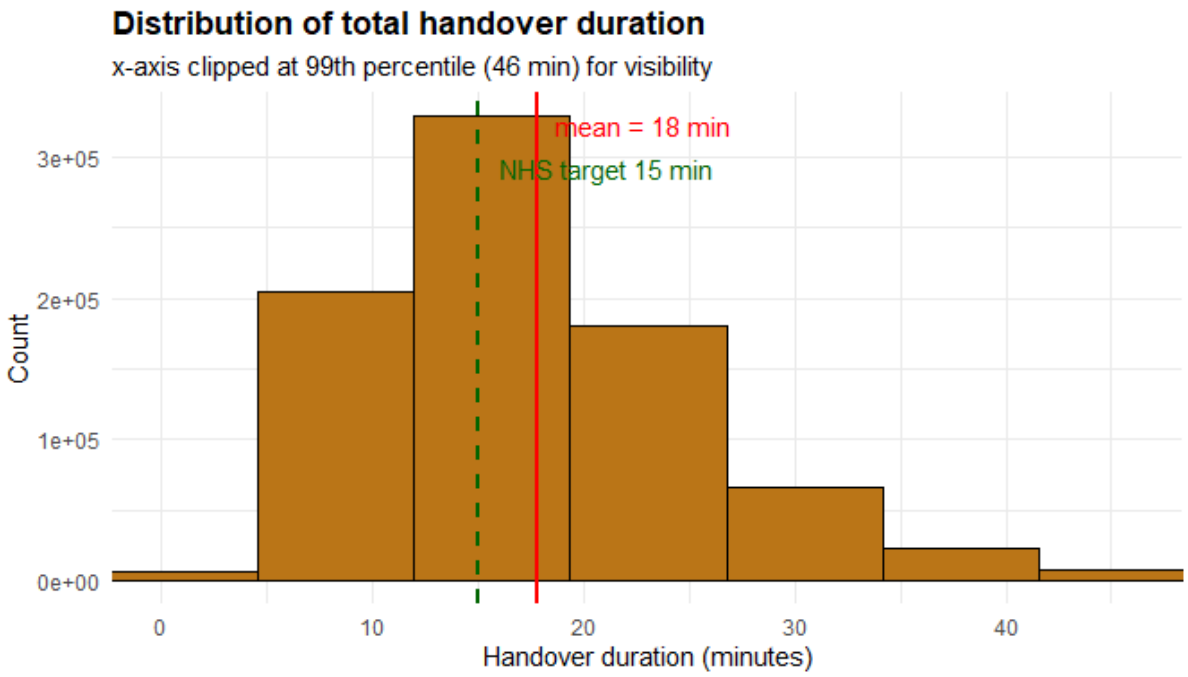


Figure A.10: Distribution of total handover duration (wait for a handover slot plus clinical handover time) across all handovers completed after the warm-up period in the baseline scenario. The green dashed line marks the 15-minute NHS handover target; the red solid line marks the simulated mean of 18 minutes, showing that even under the relatively optimistic baseline assumptions the mean exceeds the NHS target.

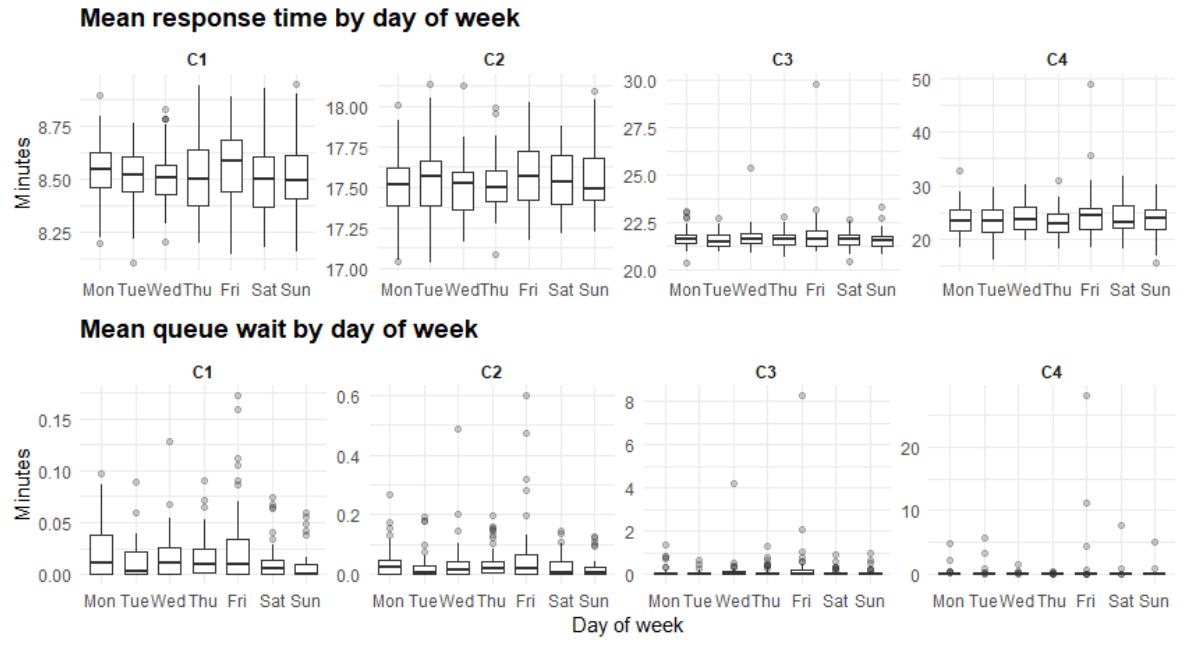


Figure A.11: Mean response time (top row) and mean call-queue wait (bottom row) by day of the week and by call category (C1–C4), under the baseline scenario. Boxplots summarise variation across 15 replications. Neither response time nor queue wait shows a clear day of week effect for any category, indicating that under the baseline the system's performance is driven primarily by within-day demand patterns rather than between-day variation.

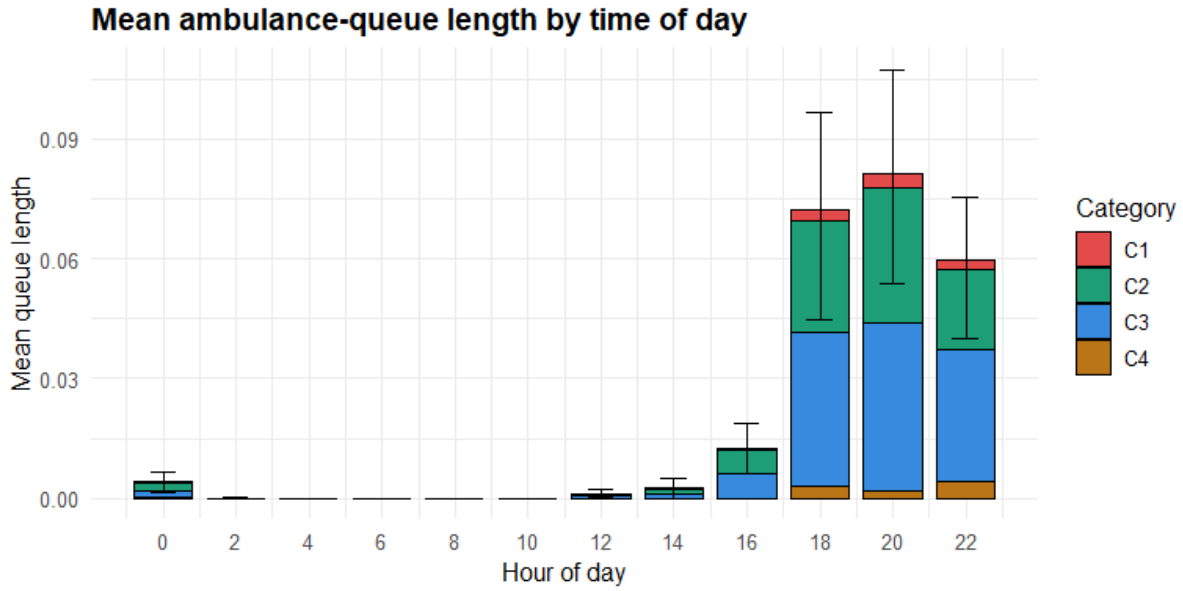


Figure A.12: Mean ambulance-queue length (calls waiting for an available ambulance) by hour of day, stacked by call category, under the baseline scenario. Error bars show 95% confidence intervals across 15 replications. The queue is negligible overnight and from early morning, building rapidly from 16:00 and peaking around 20:00 before easing overnight. The build-up is driven largely by Category 2 and Category 3 calls.

Appendix B

Logistic regression

Valenzuela et al. also motivate logistic regression as preferable to linear regression for survival probabilities, noting that linear models can generate impossible predictions (probability of survival < 0 or > 1). Logistic regression does not have this issue, yielding models with greater face validity [72].

Let $Y \in \{0, 1\}$ denote a binary outcome (e.g. survival to discharge) with

$$p(\mathbf{x}) = \mathbb{P}(Y = 1 \mid \mathbf{x}). \tag{B.1}$$

Let $\mathbf{x}_i \in \mathbb{R}^{p+1}$ denote the covariate vector for SCA i, augmented to include an intercept term, i.e. $\mathbf{x}_i = (1, x_{i1}, \ldots, x_{ip})^\top$. Let $\beta \in \mathbb{R}^{p+1}$ be the corresponding vector of regression coefficients, where the first element is the intercept.

Logistic regression links the covariates $\mathbf{x}_i$ to the probability through the log-odds (logit) transformation:

$$\log\left(\frac{p(\mathbf{x}_i)}{1 - p(\mathbf{x}_i)}\right) = \mathbf{x}_i^\top \beta. \tag{B.2}$$

Rearranging gives the logistic model function:

$$p(\mathbf{x}_i) = \frac{1}{1 + \exp\left(-\mathbf{x}_i^\top \beta\right)}. \tag{B.3}$$

[110]

Parameters are typically estimated by maximum likelihood where exponentiating coefficients gives odds ratios, which is often a convenient interpretation in applied health modelling [110].

In Valenzuela et al. (1997), model performance was assessed using area under the ROC curve and calibrated via an observed-vs-expected comparison based on the Hosmer–Lemeshow goodness-of-fit approach [72, 111].

Following Cairns et al. (2011) [1], the simulation treats short-term survival as primarily driven by whether the arrest is witnessed with an initial VF rhythm, and then by the delays to CPR and defibrillation.

For cases simulated as witnessed VF, survival to hospital discharge is generated using the simplified Valenzuela logistic survival model based only on:

- I_{CPR}: interval (minutes) from collapse to CPR initiation,
- I_{defib}: interval (minutes) from collapse to defibrillation initiation.

Let t_{resp} denote the simulated response time (minutes). Then

$$I_{\text{defib}} = t_{\text{resp}} + 1, \qquad I_{\text{CPR}} = t_{\text{resp}} + 2. \tag{B.4}$$

where the additional 1–2 minutes represent practical on-scene delays to deliver the first intervention after arrival (positioning, attaching AED, initiating compressions).

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